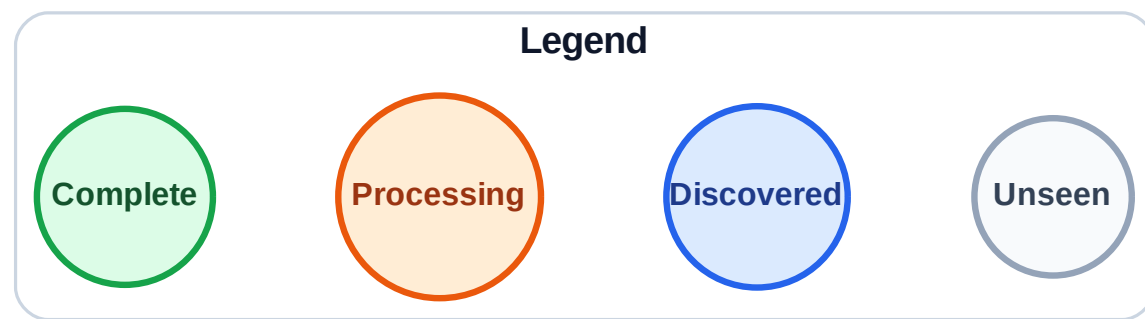


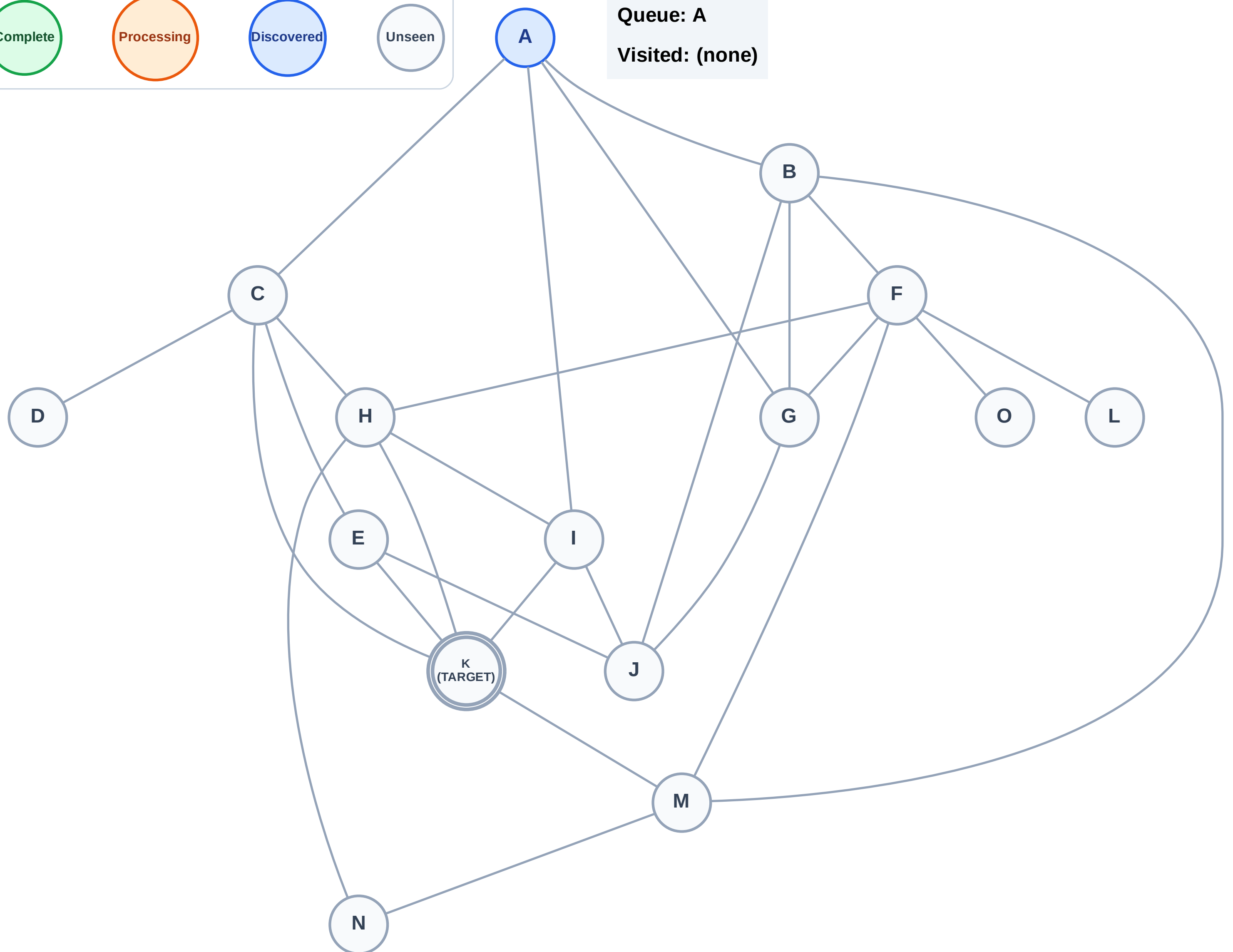
BFS Traversal

Step 1: Start at A (Seeking target node K)



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

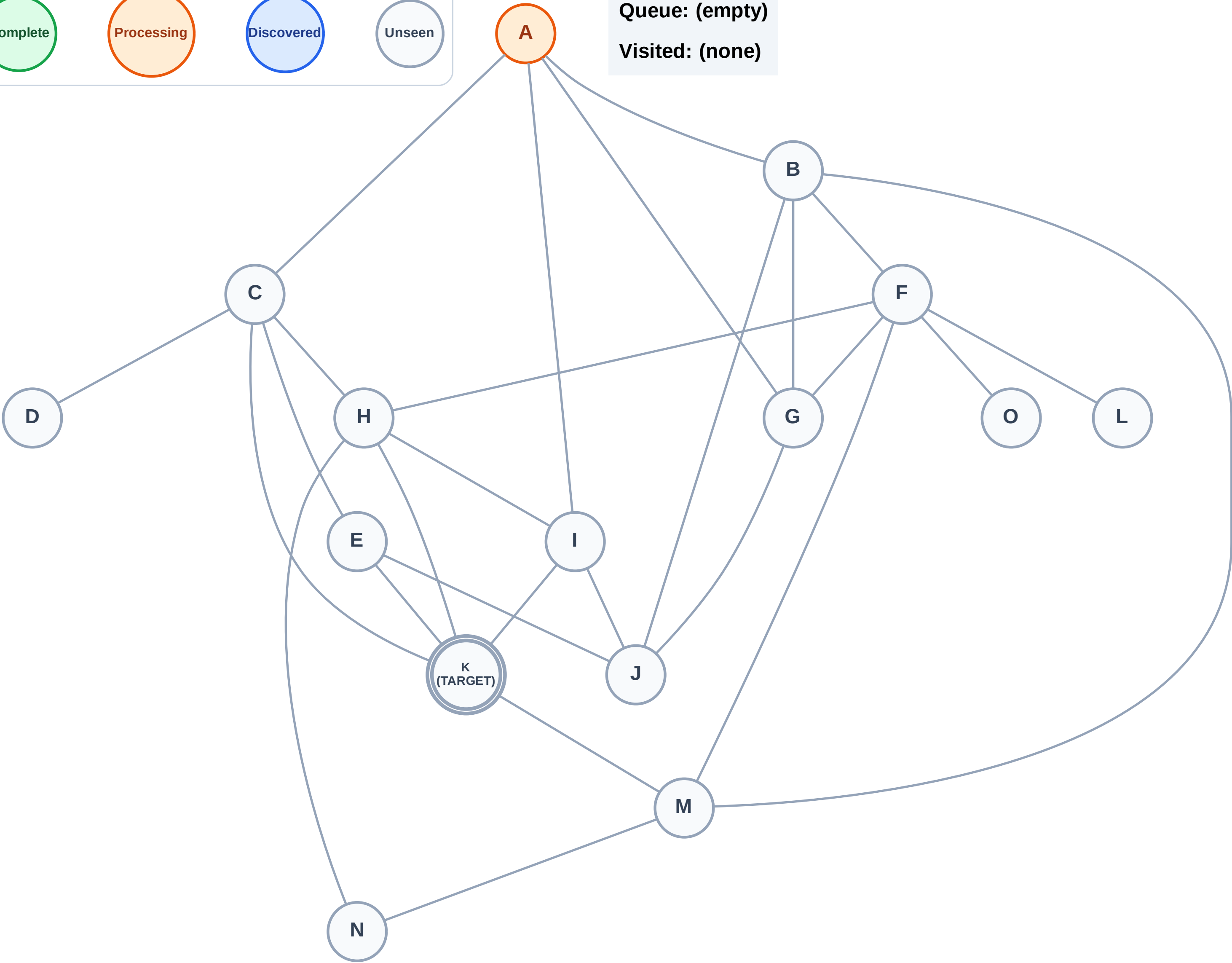
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

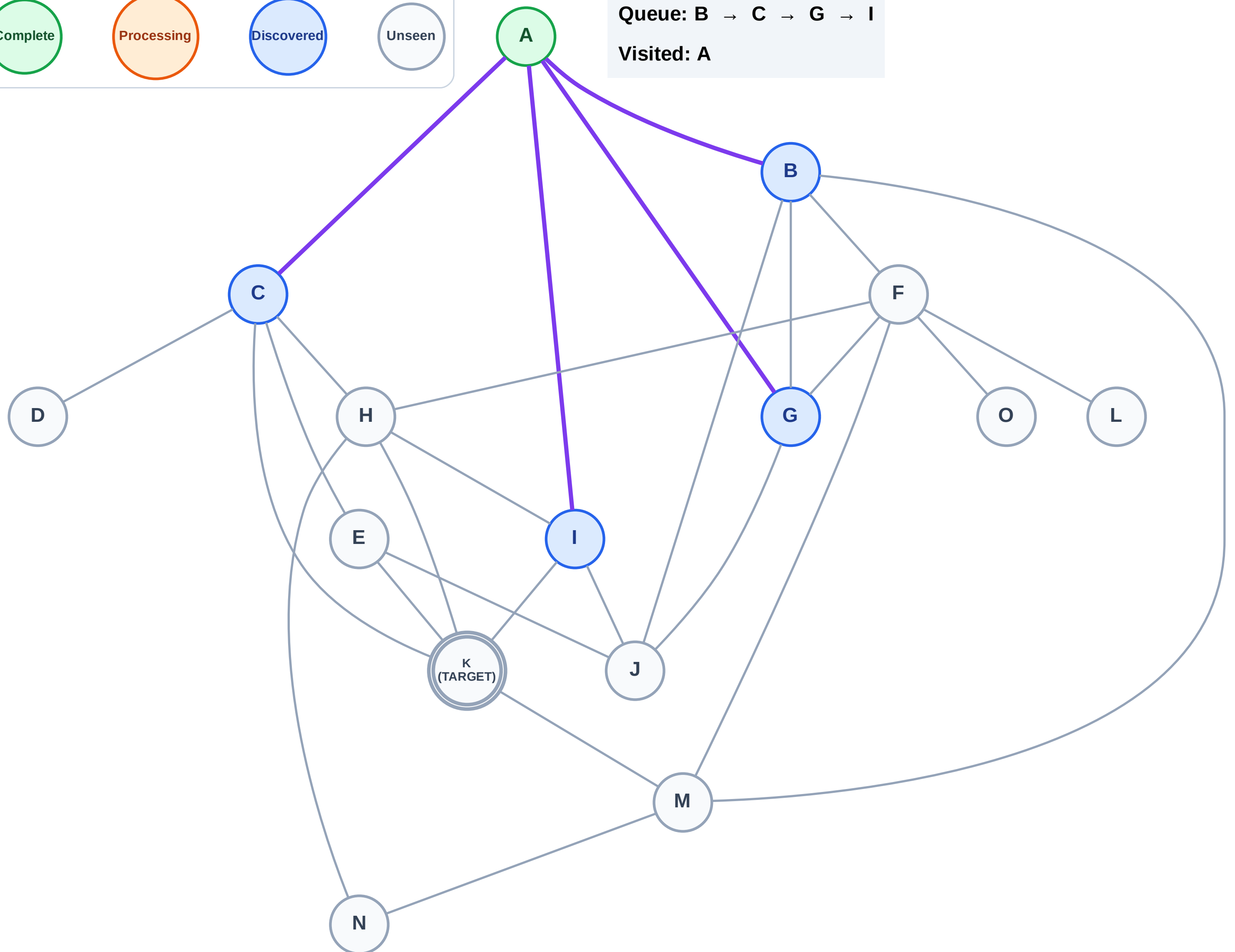
Step 3: Discover B, C, G, I from A

Legend



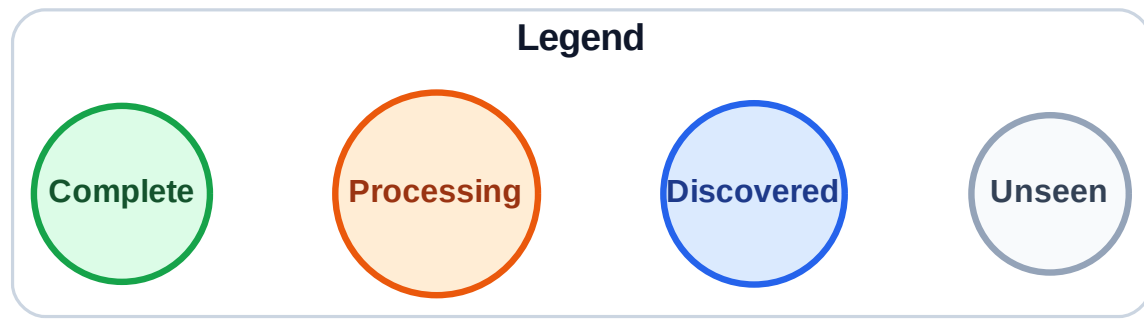
Queue: B → C → G → I

Visited: A



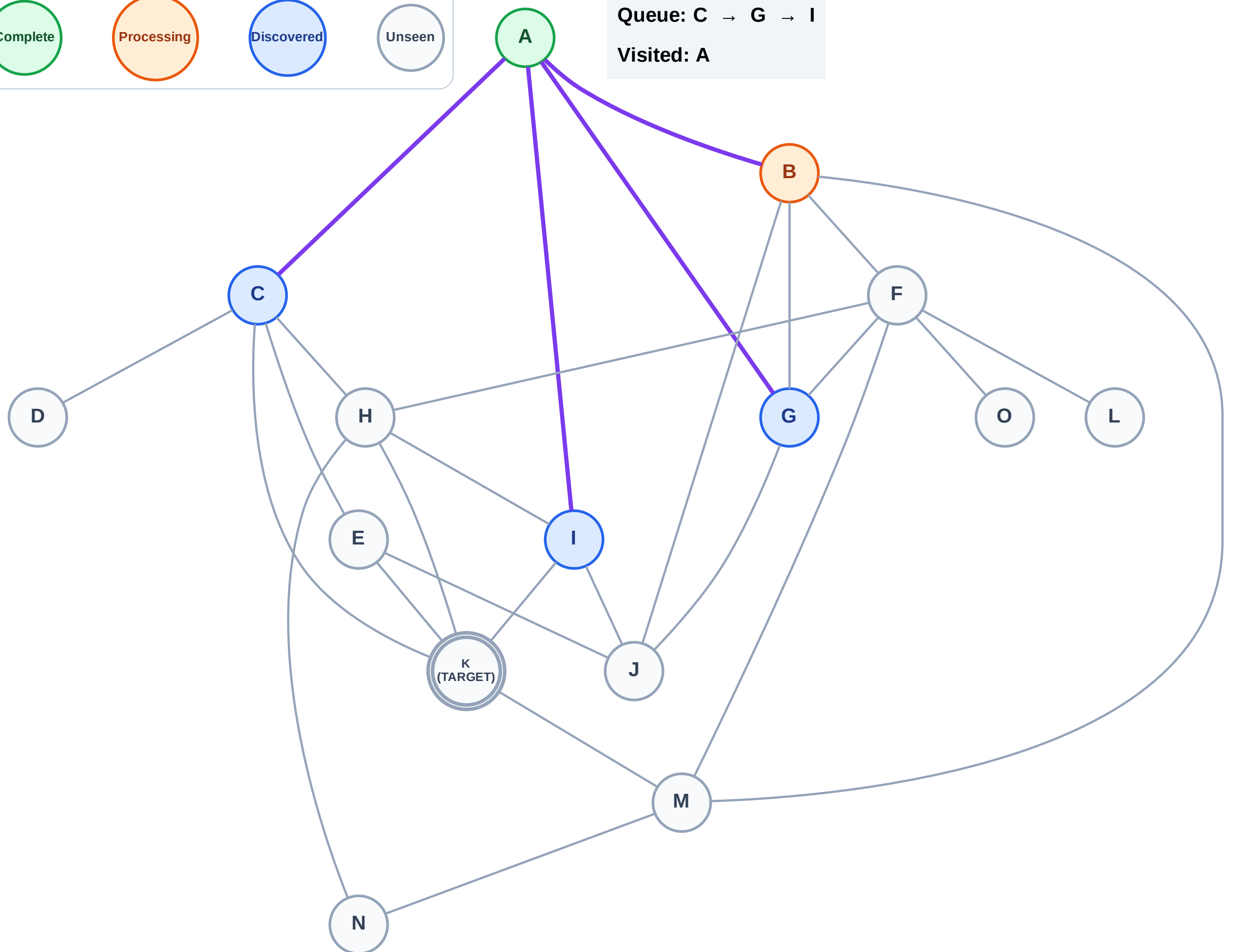
BFS Traversal

Step 4: Process node B



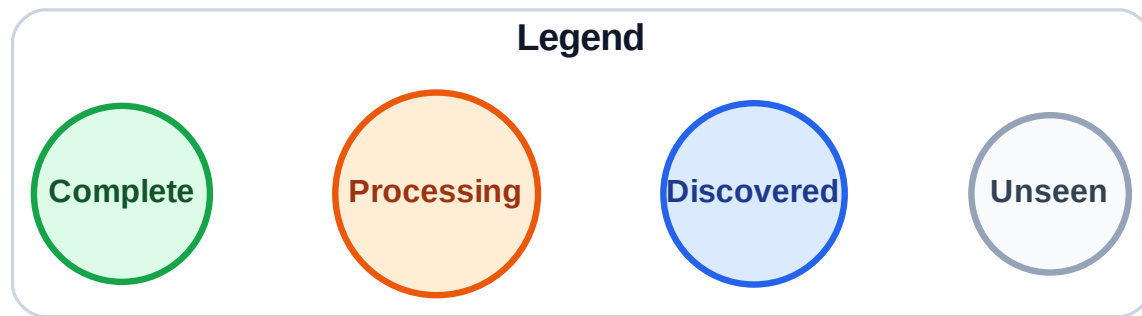
Queue: C → G → I

Visited: A



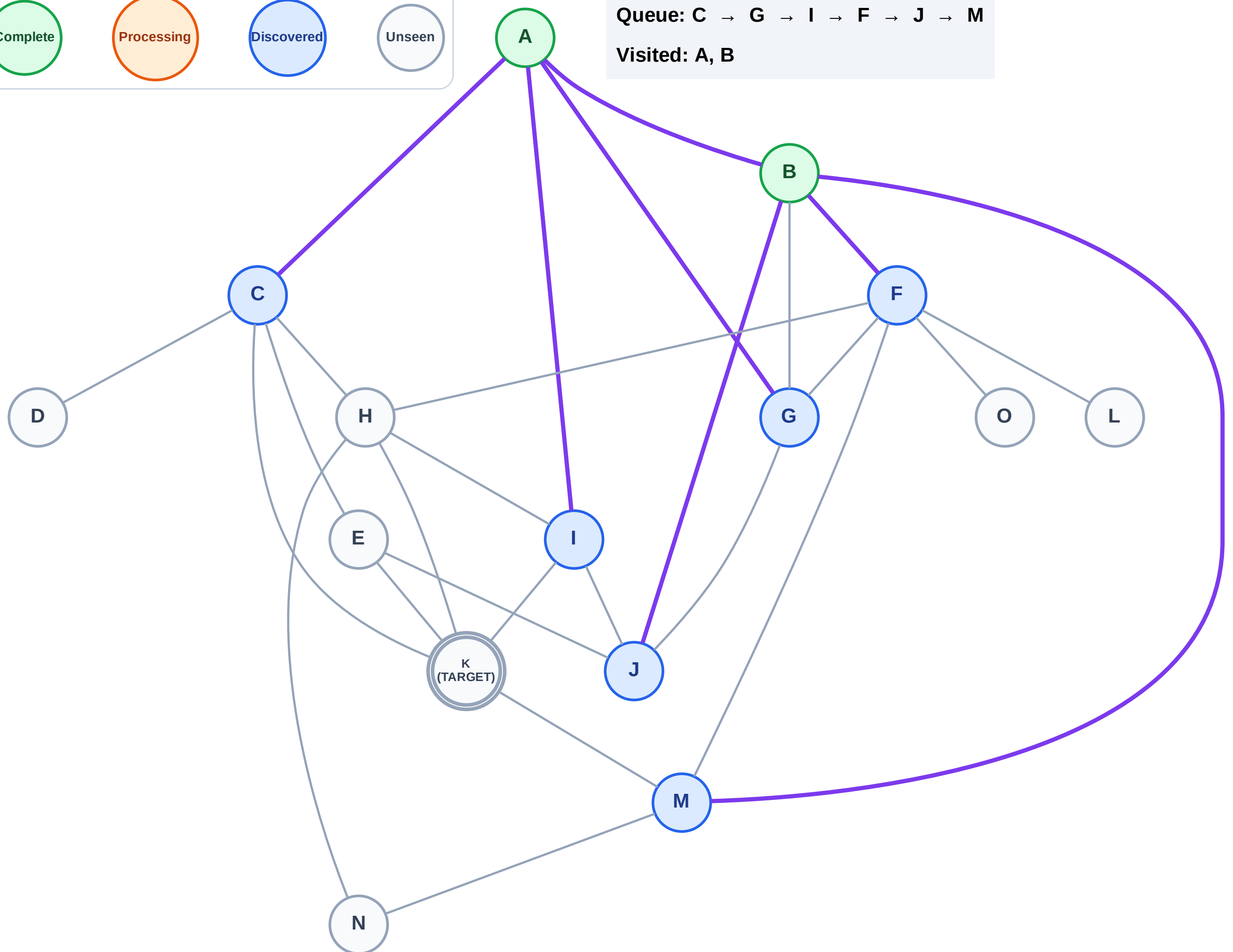
BFS Traversal

Step 5: Discover F, J, M from B



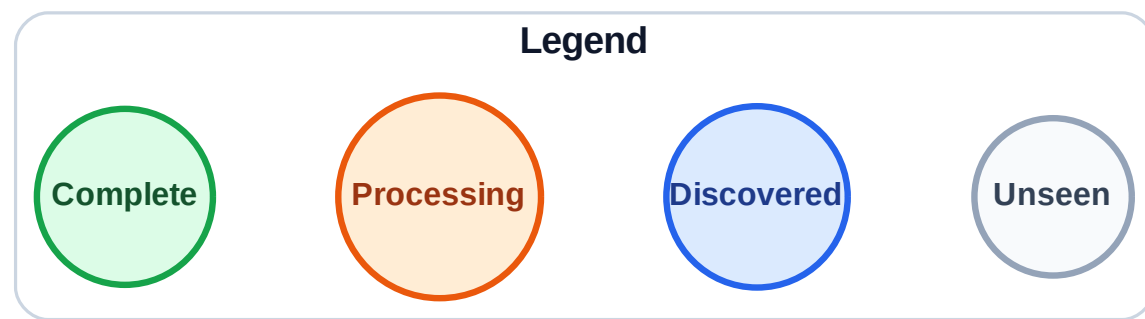
Queue: C → G → I → F → J → M

Visited: A, B



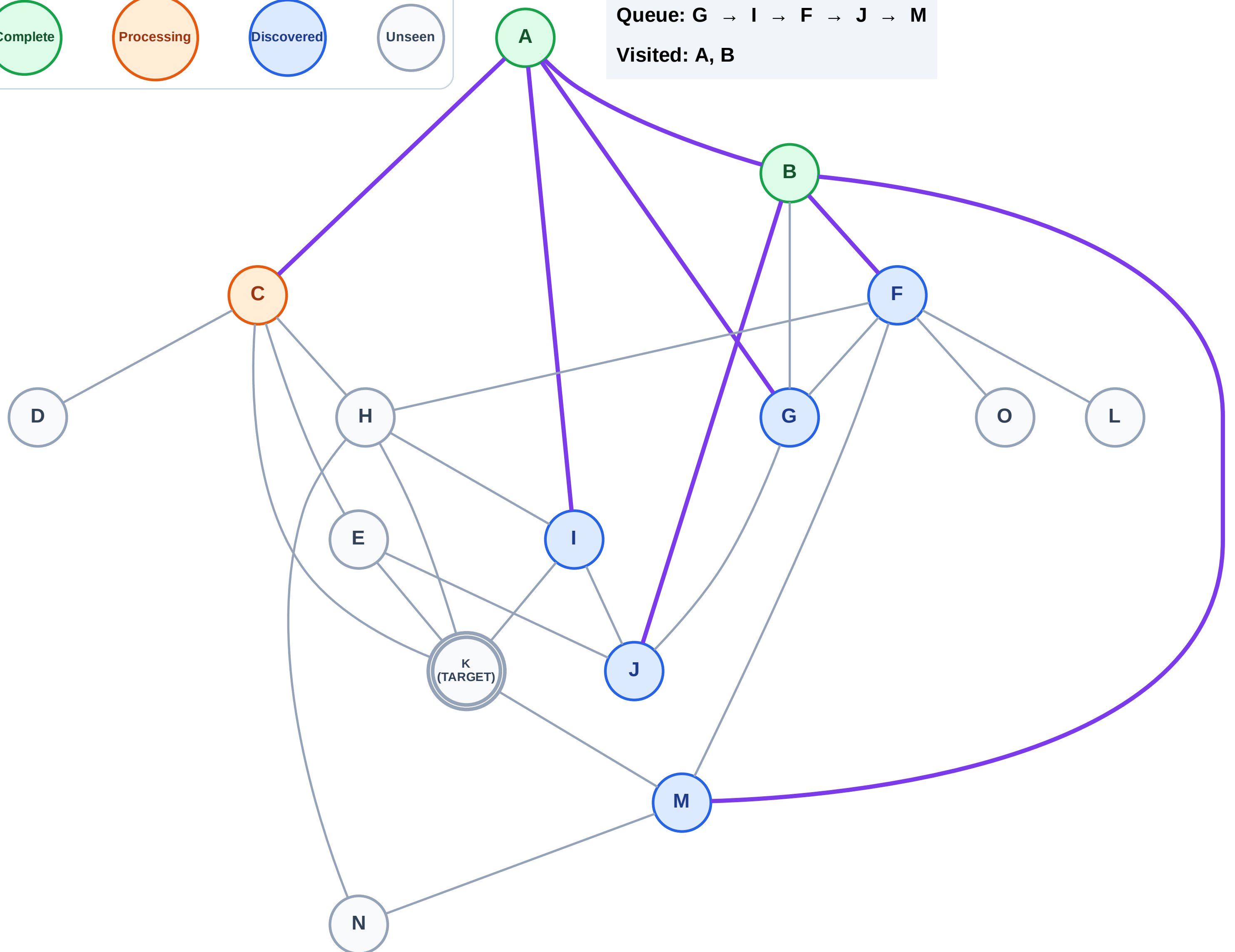
BFS Traversal

Step 6: Process node C



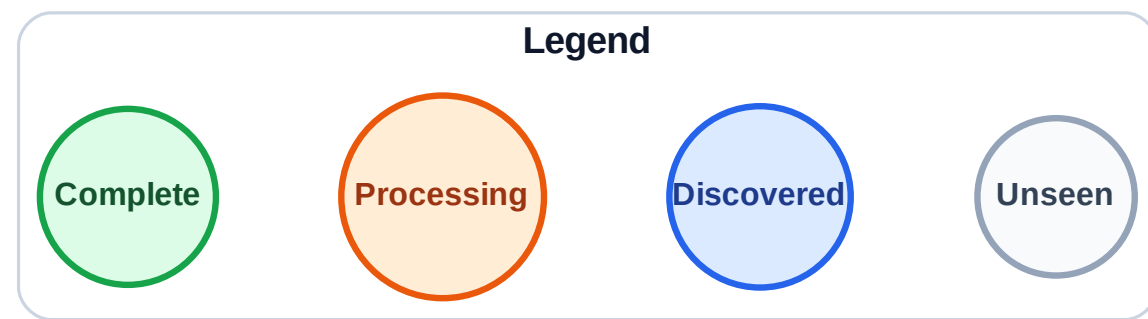
Queue: G → I → F → J → M

Visited: A, B



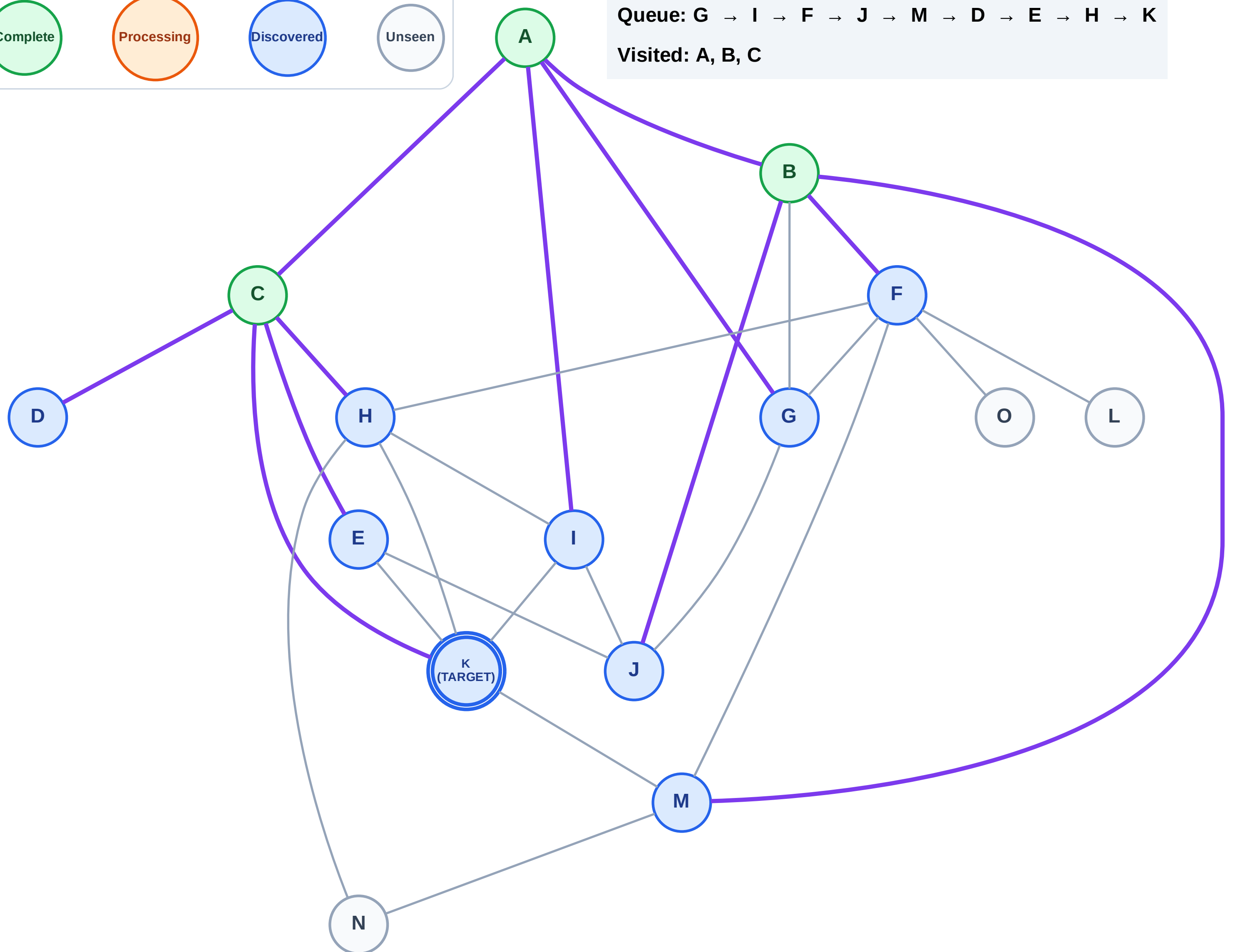
BFS Traversal

Step 7: Discover D, E, H, K from C



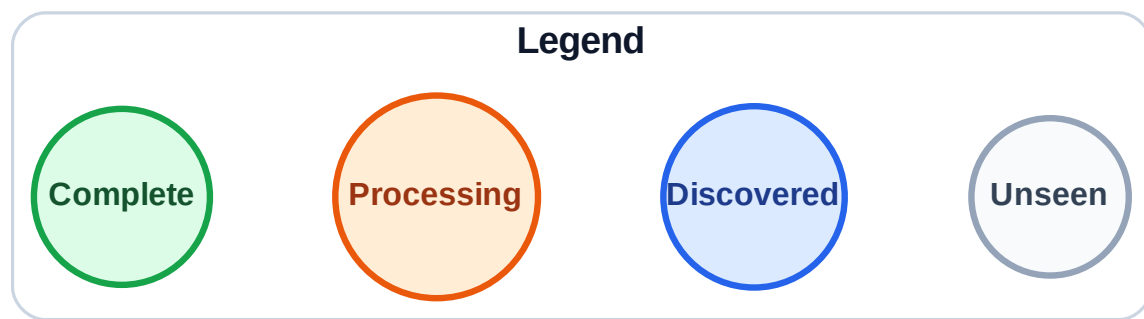
Queue: G → I → F → J → M → D → E → H → K

Visited: A, B, C



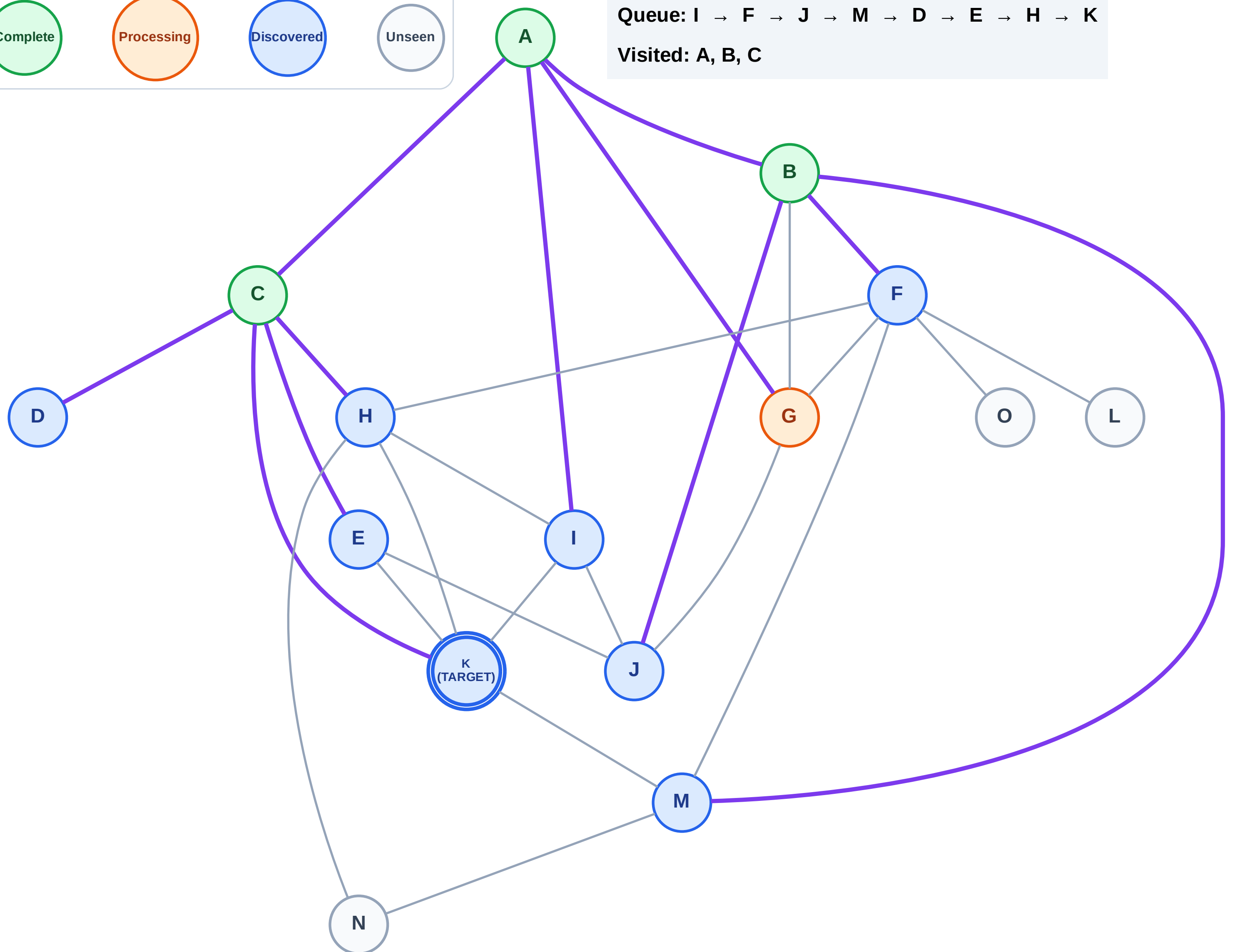
BFS Traversal

Step 8: Process node G



Queue: I → F → J → M → D → E → H → K

Visited: A, B, C



BFS Traversal

Step 9: No new neighbors from G

Legend



Complete



Processing



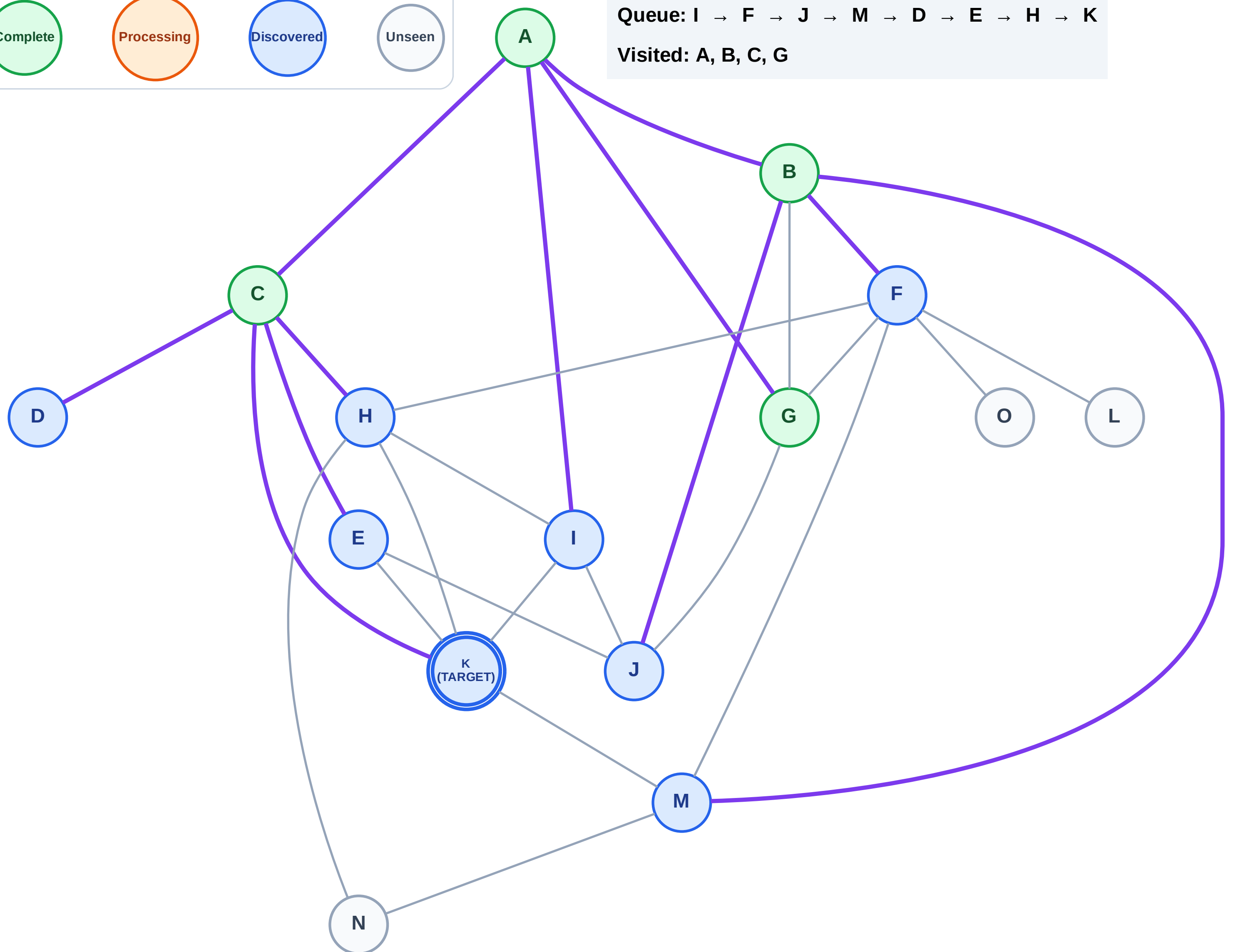
Discovered



Unseen

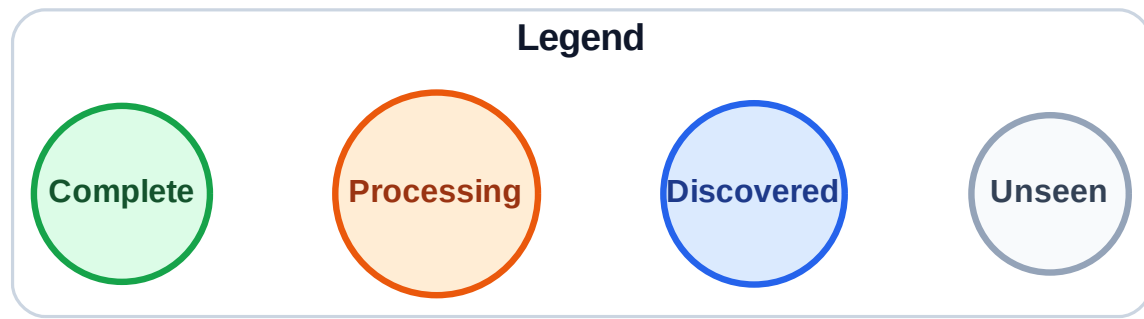
Queue: I → F → J → M → D → E → H → K

Visited: A, B, C, G



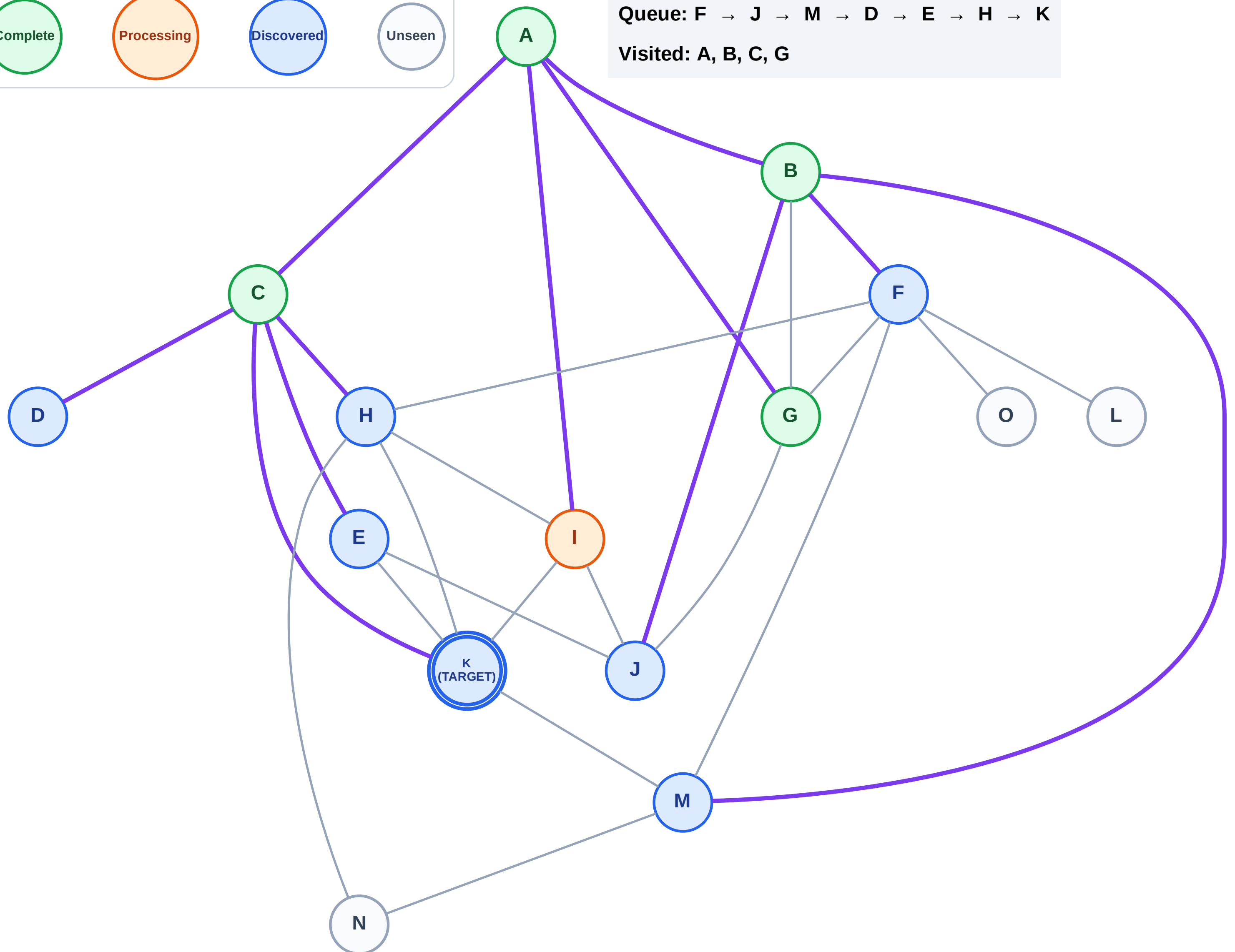
BFS Traversal

Step 10: Process node I



Queue: F → J → M → D → E → H → K

Visited: A, B, C, G



BFS Traversal

Step 11: No new neighbors from I

Legend



Complete



Processing



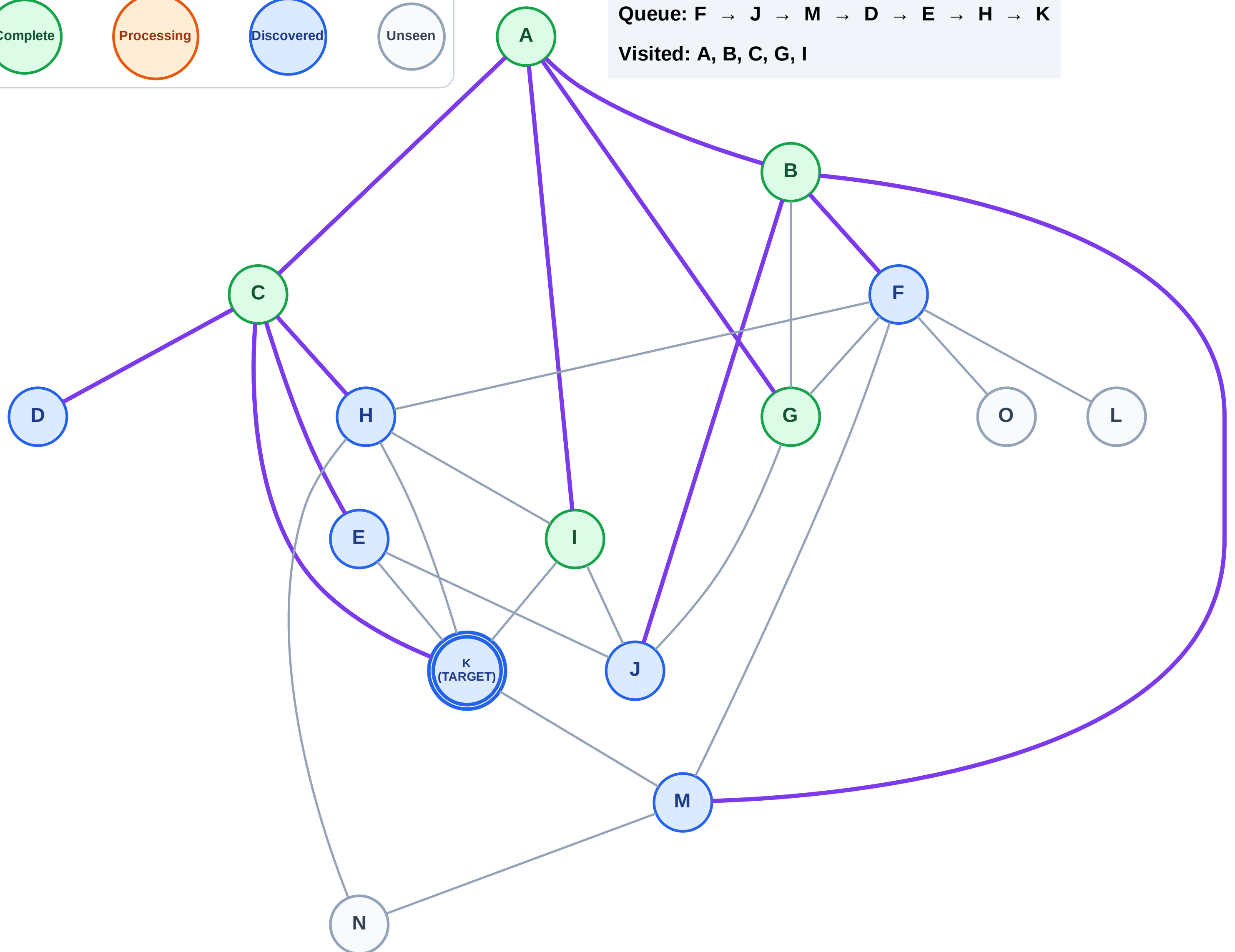
Discovered



Unseen

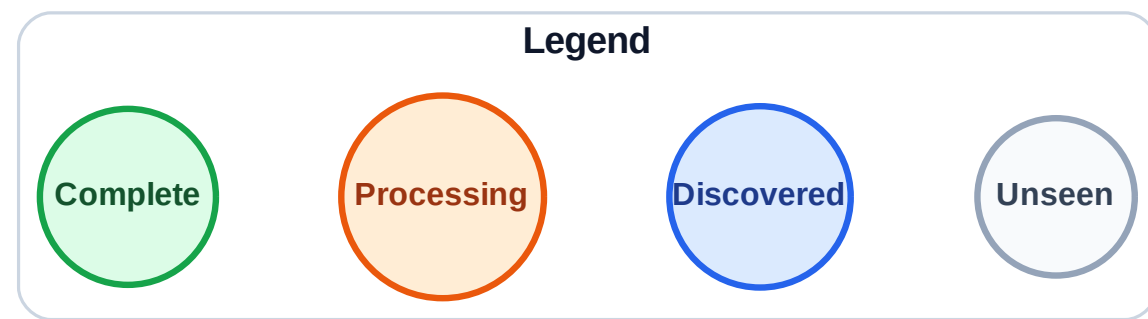
Queue: F → J → M → D → E → H → K

Visited: A, B, C, G, I



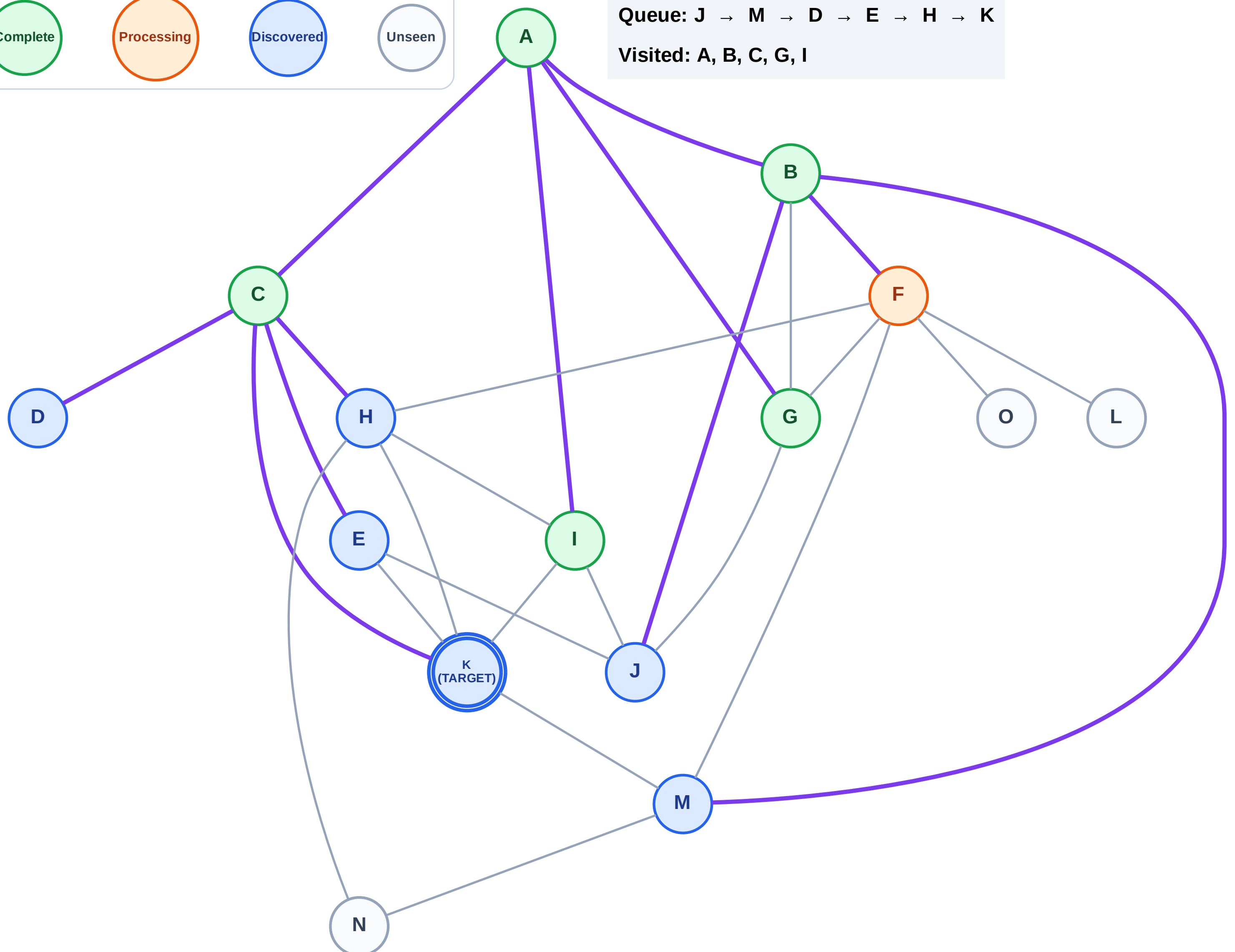
BFS Traversal

Step 12: Process node F



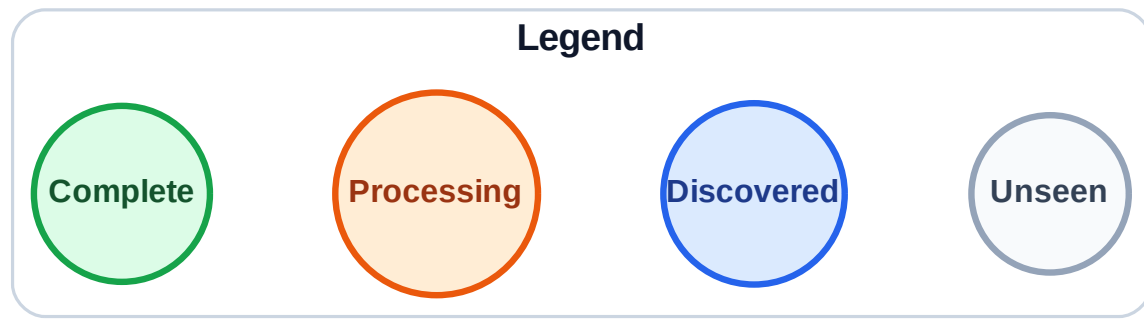
Queue: J → M → D → E → H → K

Visited: A, B, C, G, I



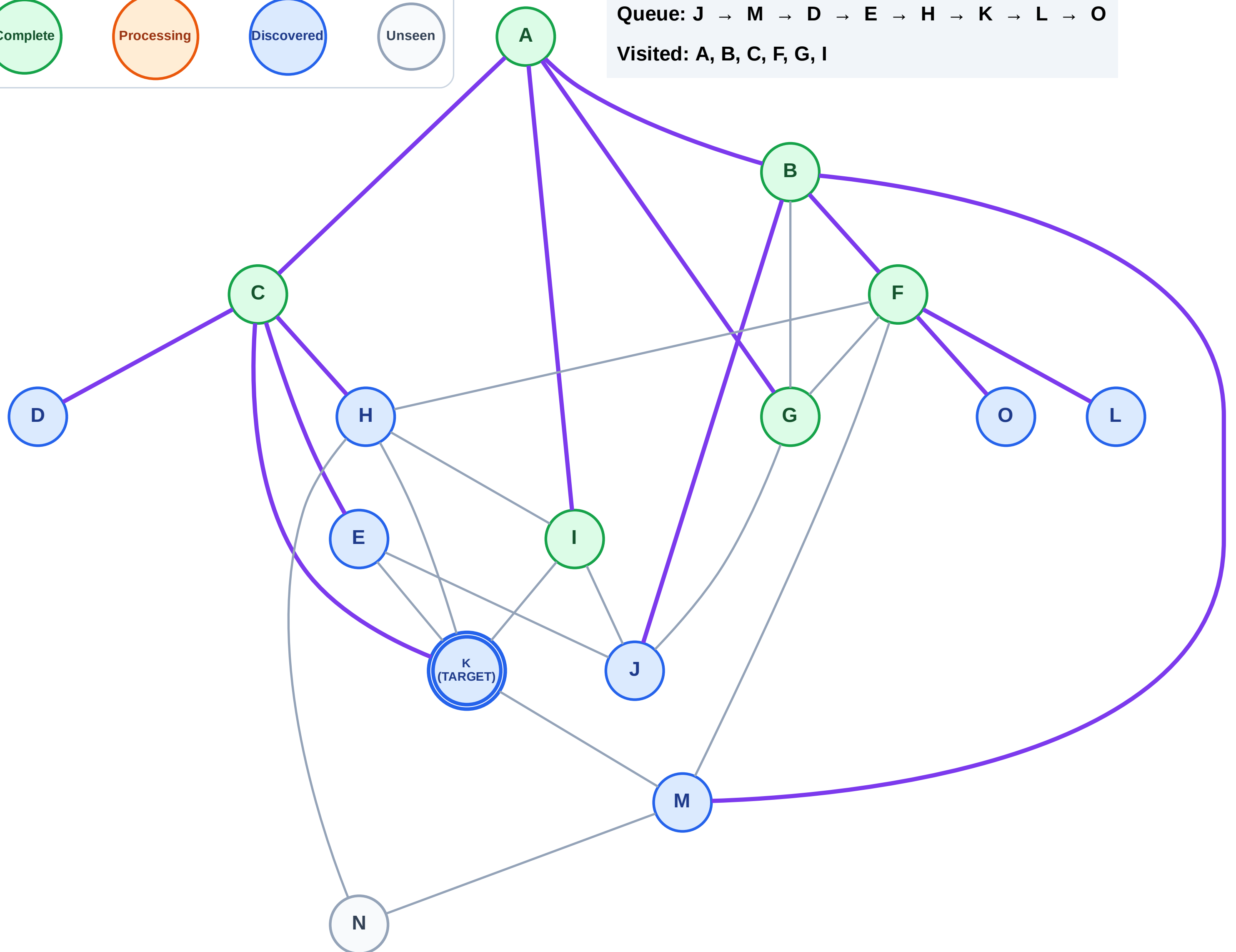
BFS Traversal

Step 13: Discover L, O from F



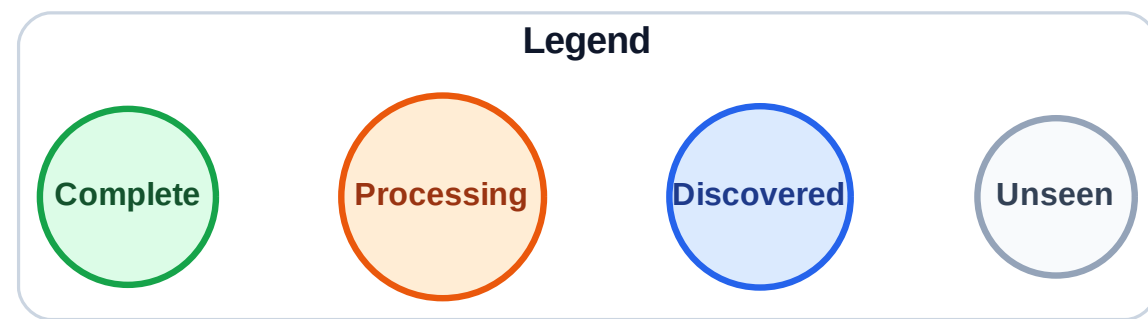
Queue: J → M → D → E → H → K → L → O

Visited: A, B, C, F, G, I



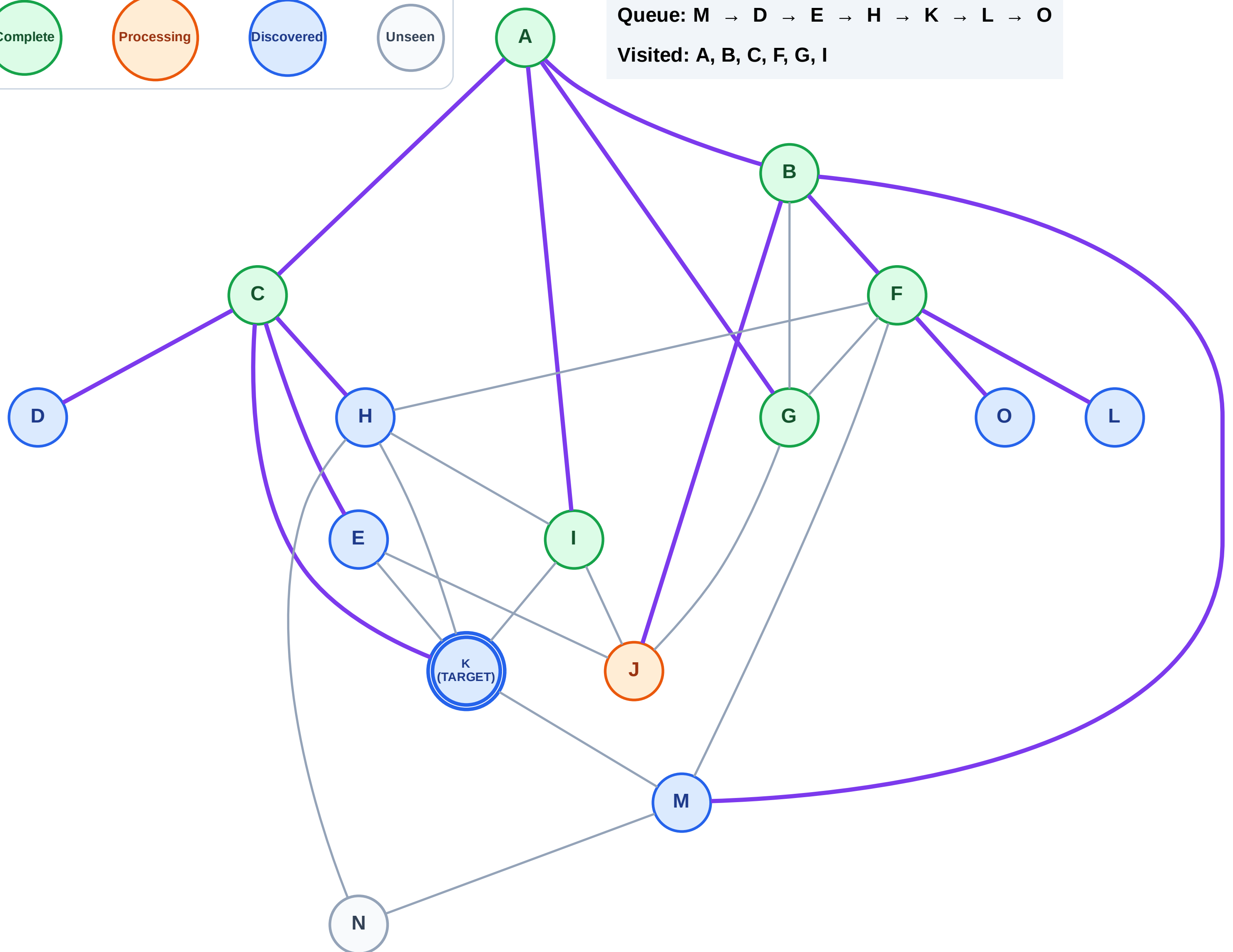
BFS Traversal

Step 14: Process node J



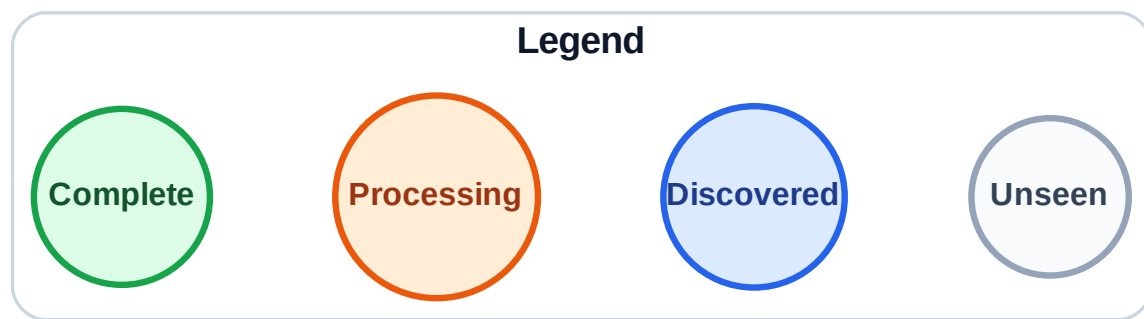
Queue: M → D → E → H → K → L → O

Visited: A, B, C, F, G, I



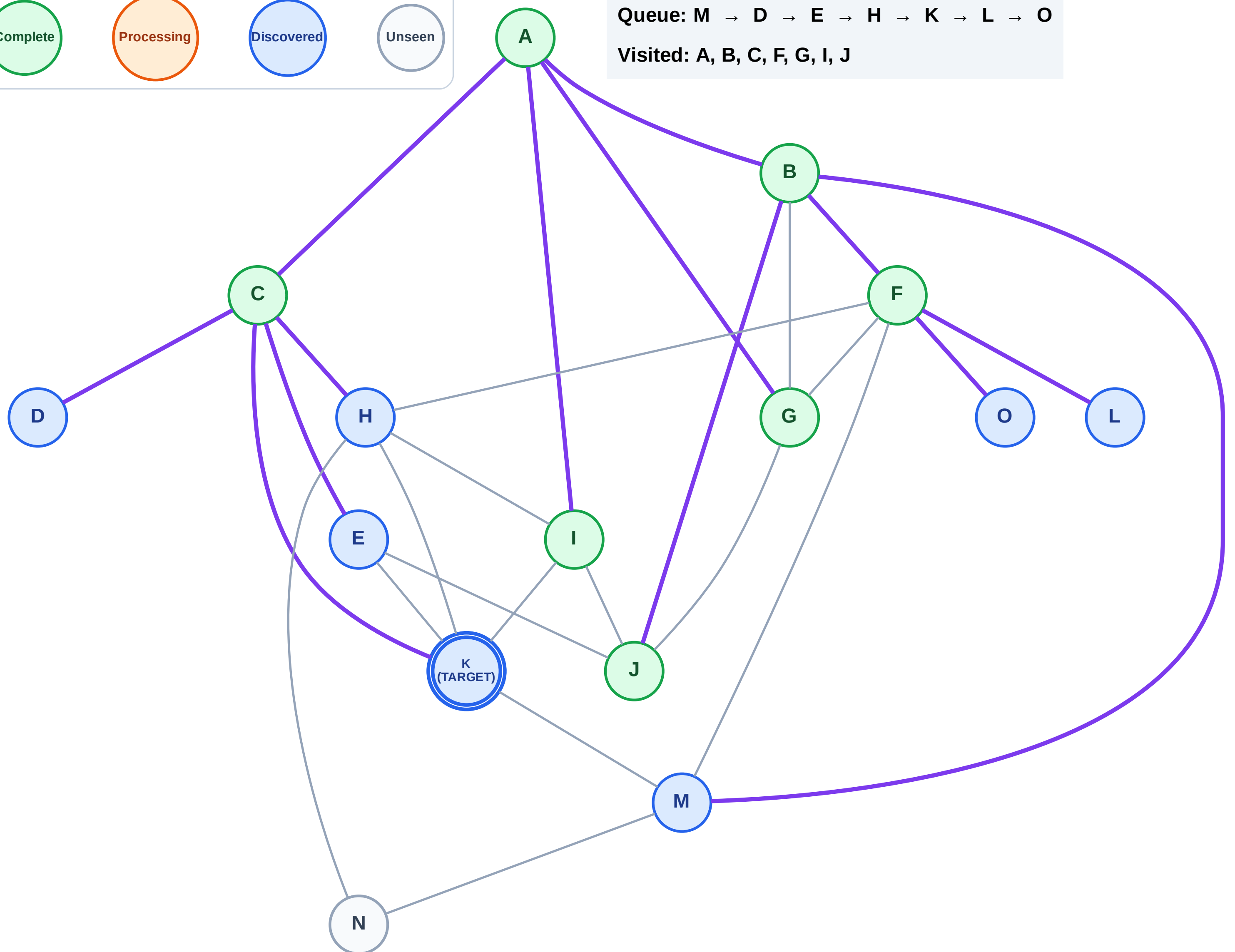
BFS Traversal

Step 15: No new neighbors from J



Queue: M → D → E → H → K → L → O

Visited: A, B, C, F, G, I, J



BFS Traversal

Step 16: Process node M

Legend

Complete

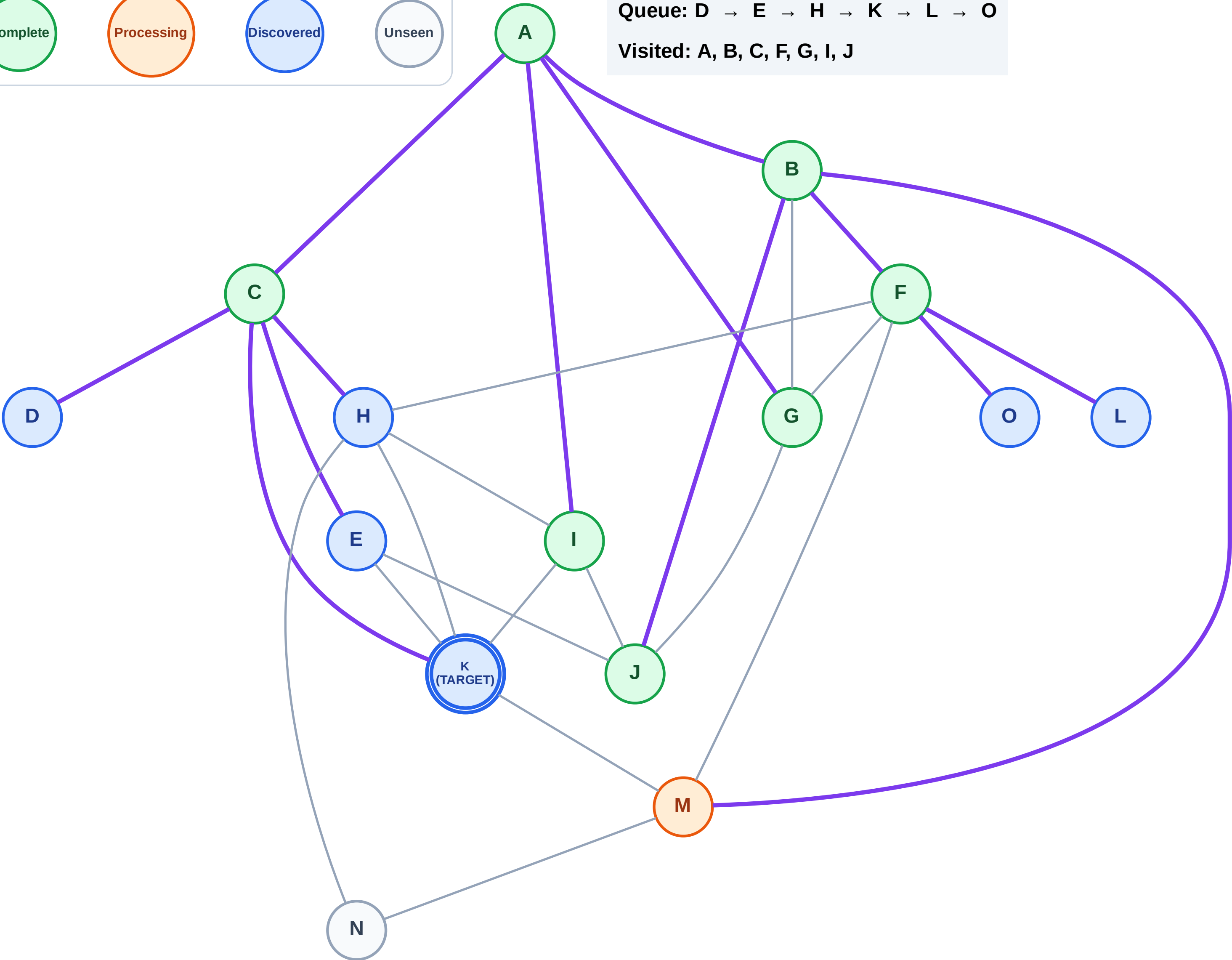
Processing

Discovered

Unseen

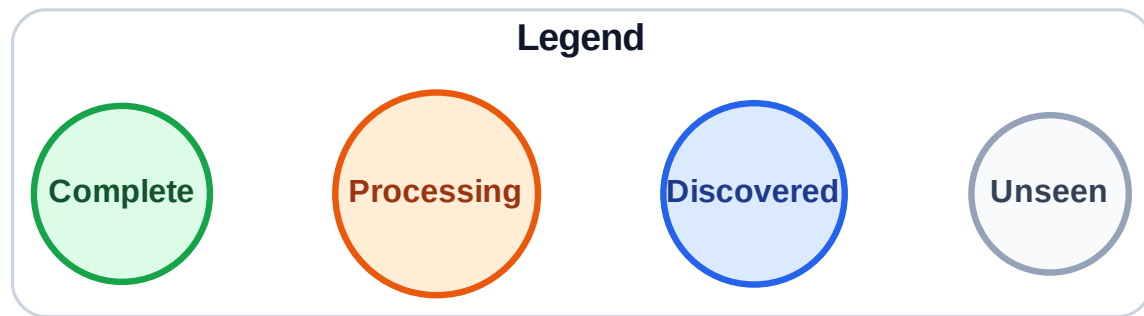
Queue: D → E → H → K → L → O

Visited: A, B, C, F, G, I, J



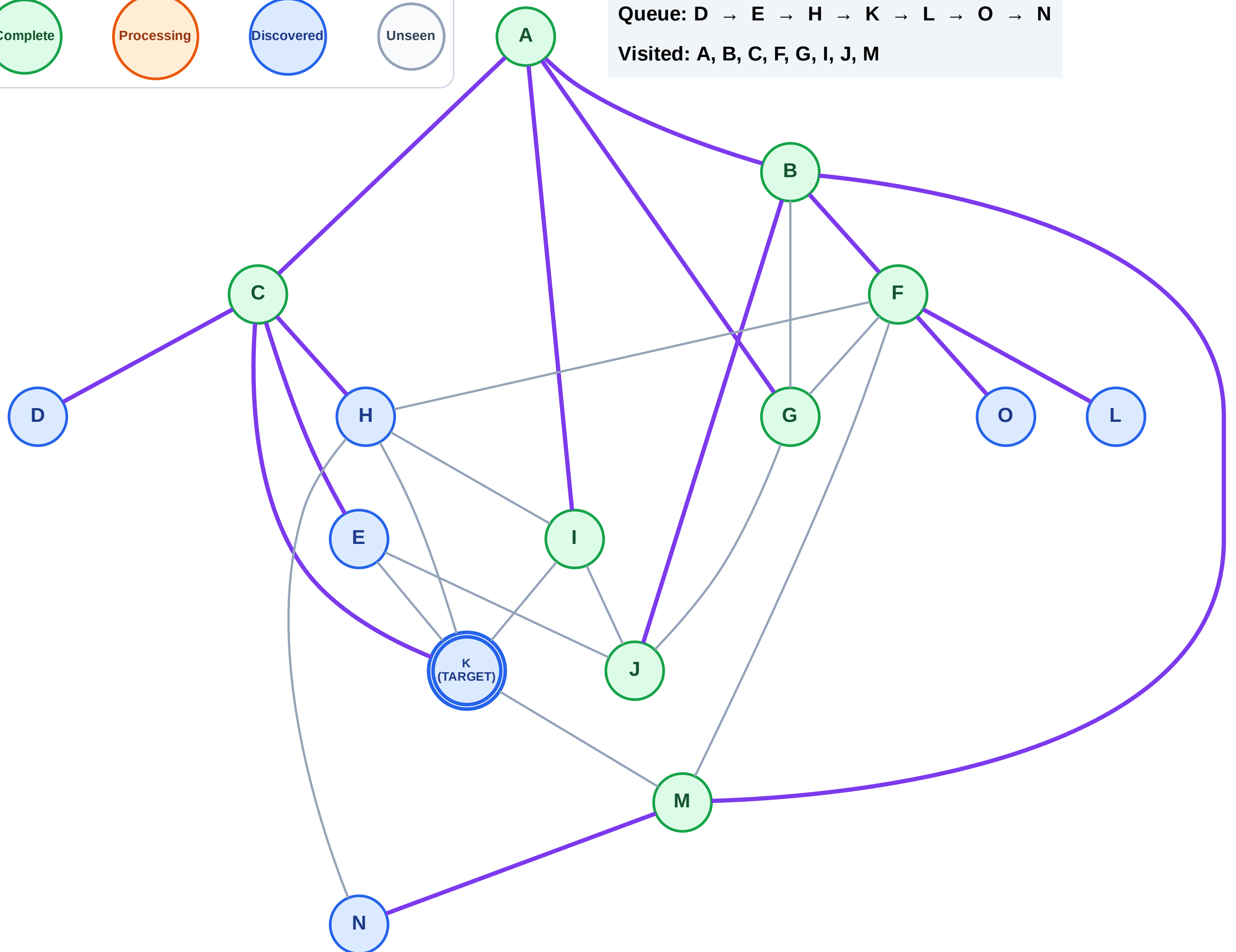
BFS Traversal

Step 17: Discover N from M



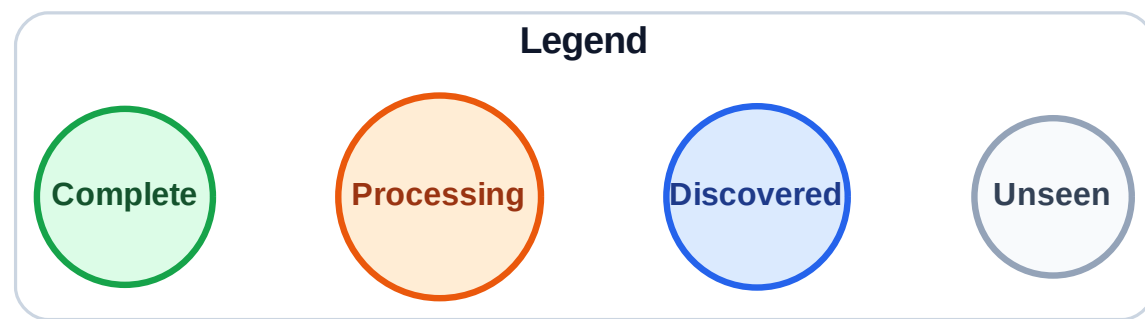
Queue: D → E → H → K → L → O → N

Visited: A, B, C, F, G, I, J, M



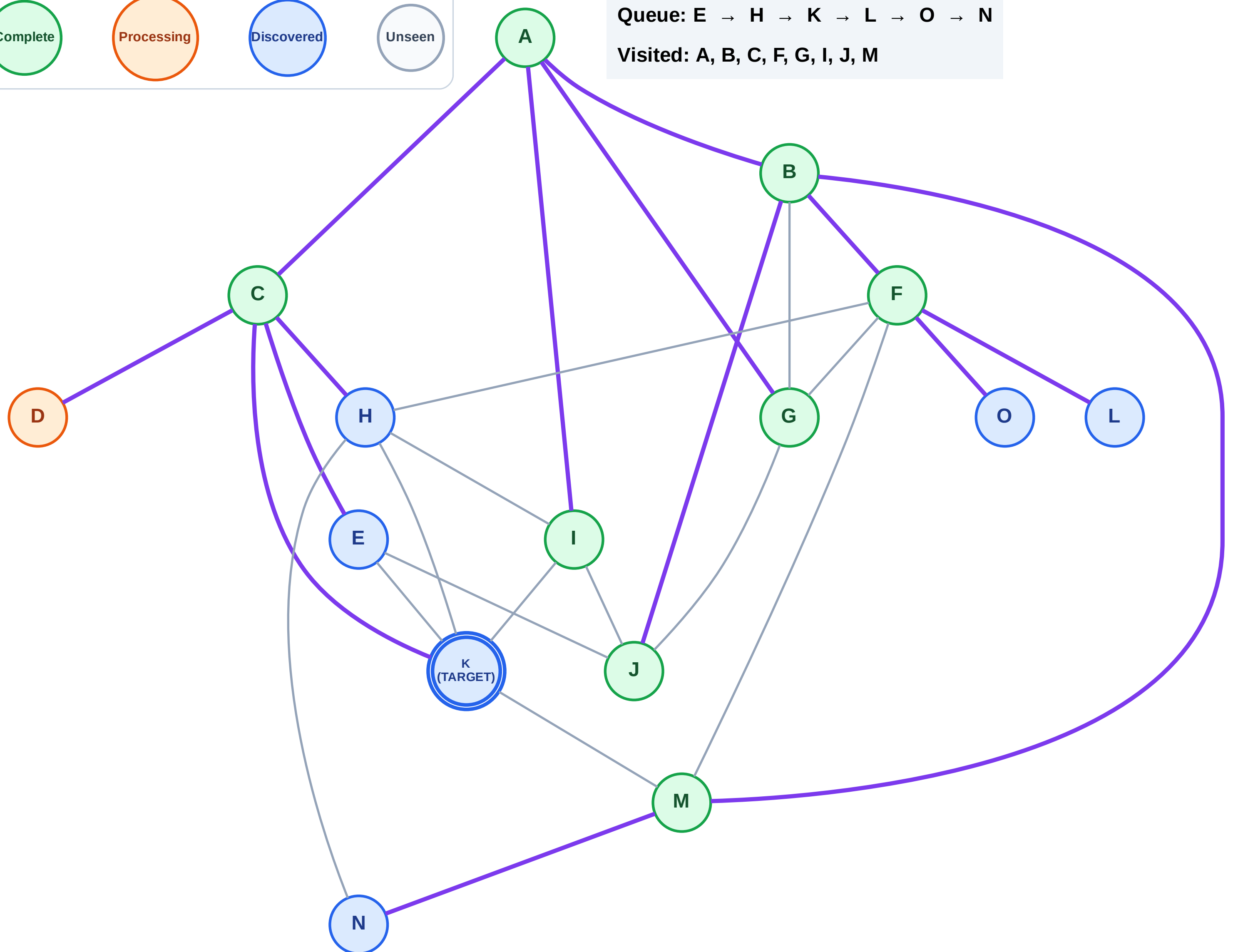
BFS Traversal

Step 18: Process node D



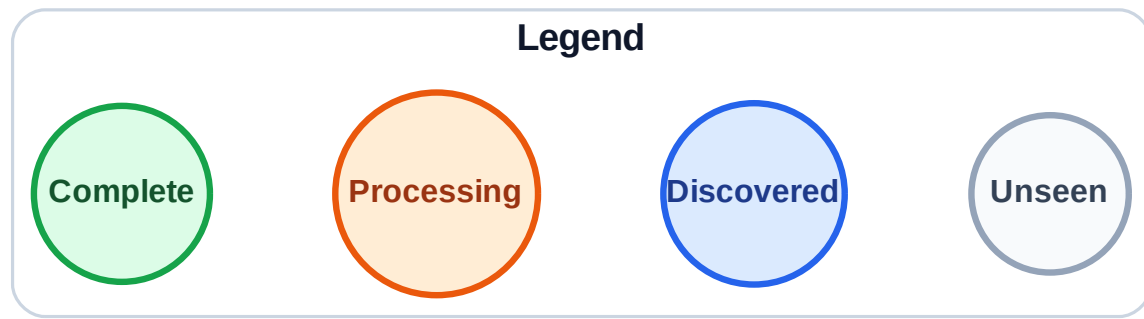
Queue: E → H → K → L → O → N

Visited: A, B, C, F, G, I, J, M



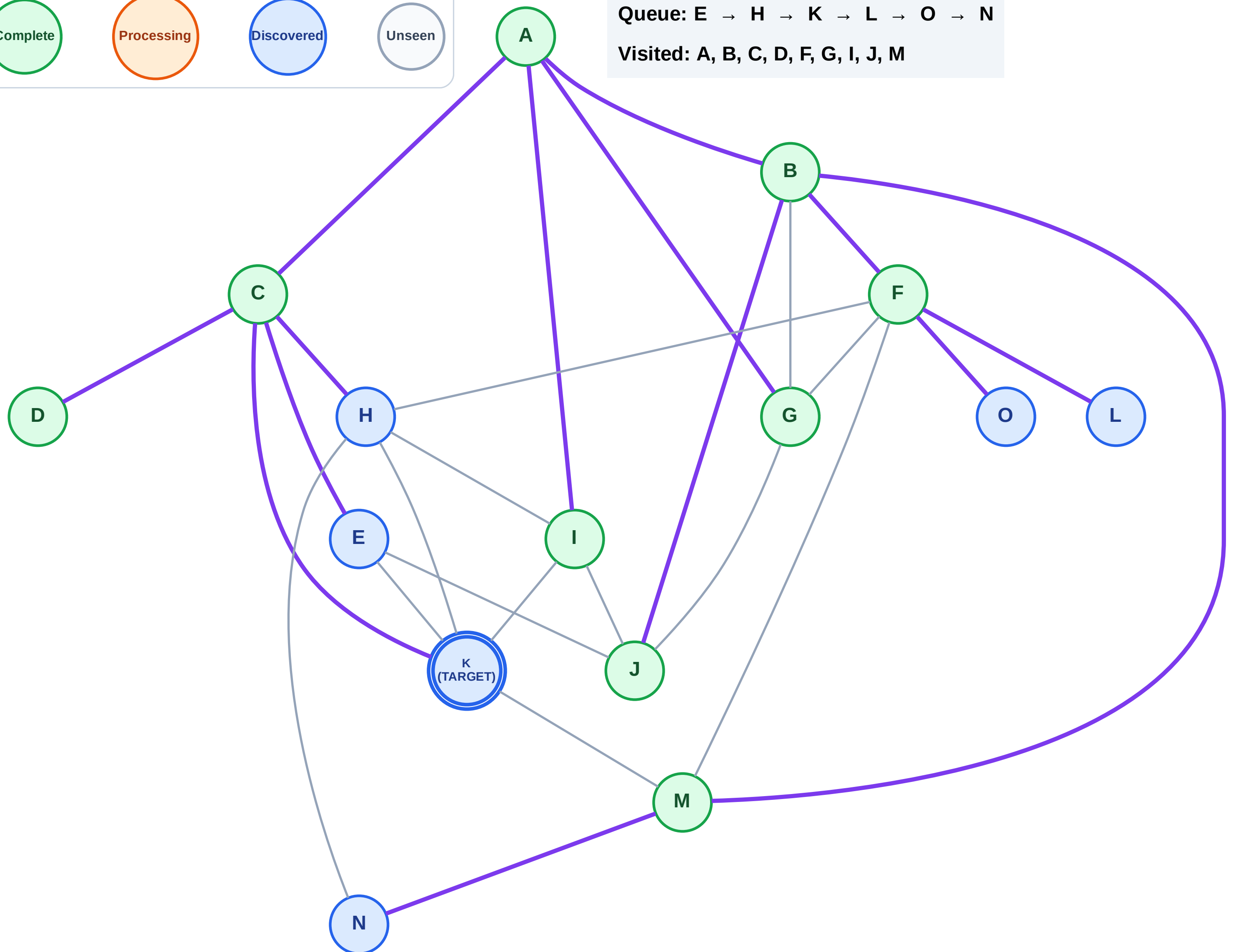
BFS Traversal

Step 19: No new neighbors from D



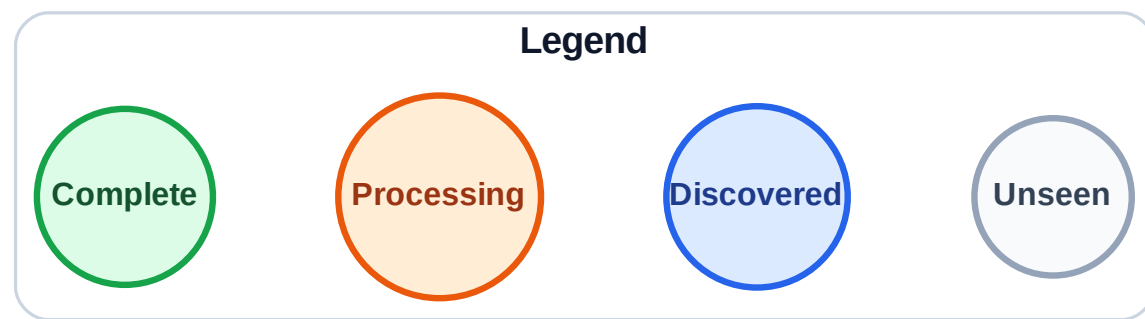
Queue: E → H → K → L → O → N

Visited: A, B, C, D, F, G, I, J, M



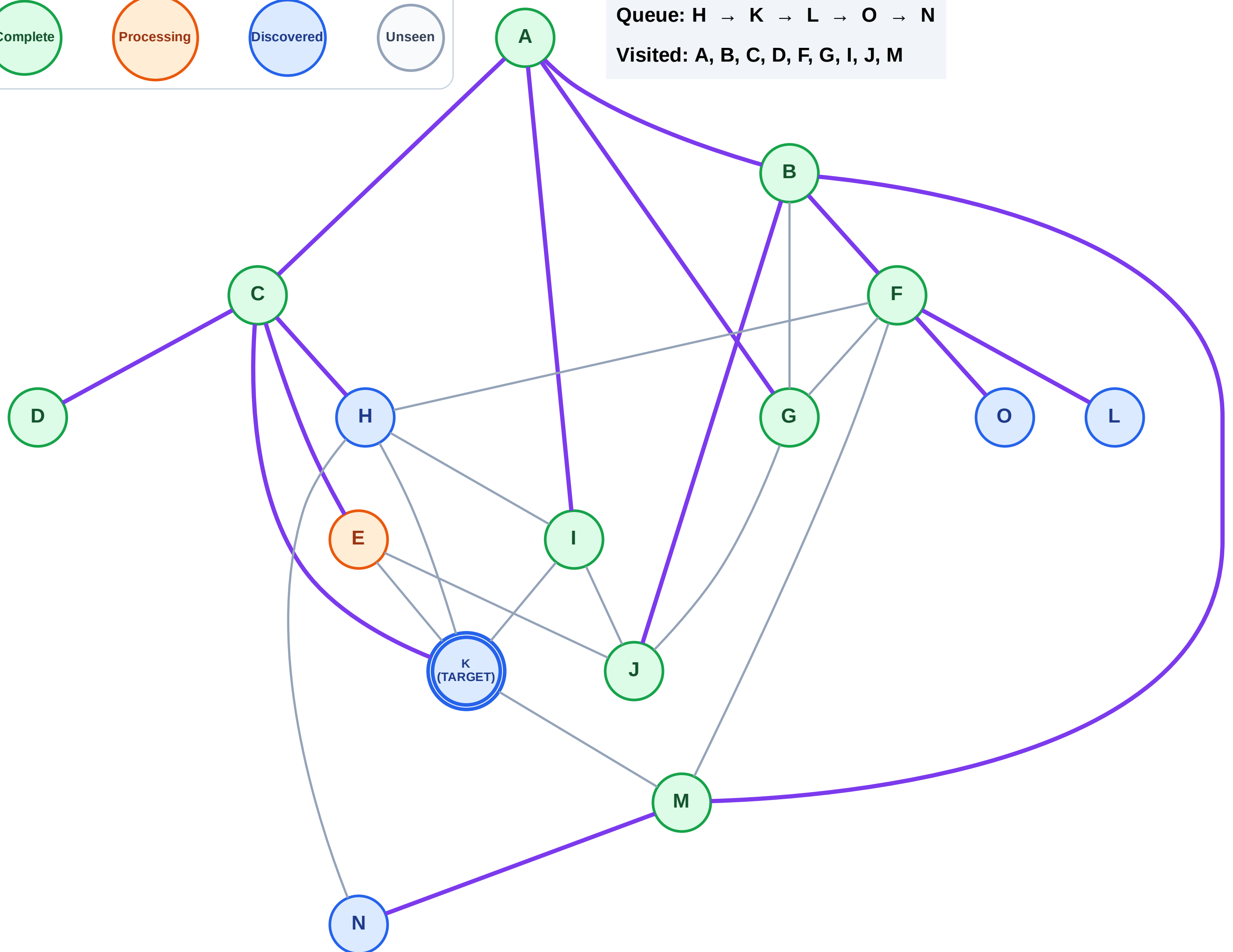
BFS Traversal

Step 20: Process node E



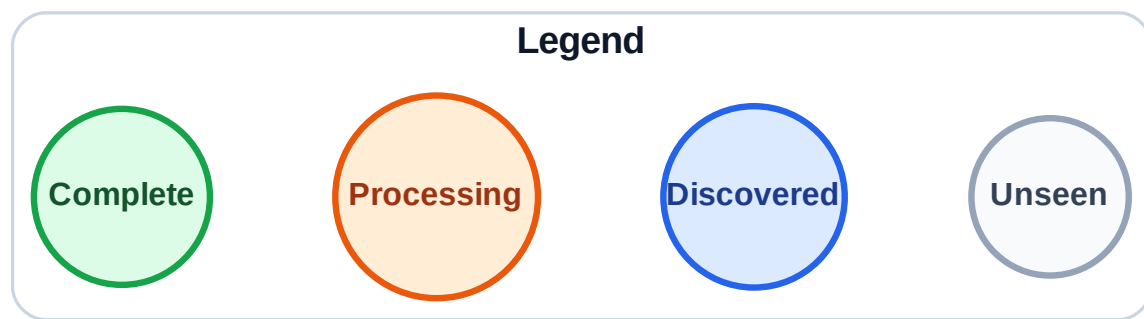
Queue: H → K → L → O → N

Visited: A, B, C, D, F, G, I, J, M



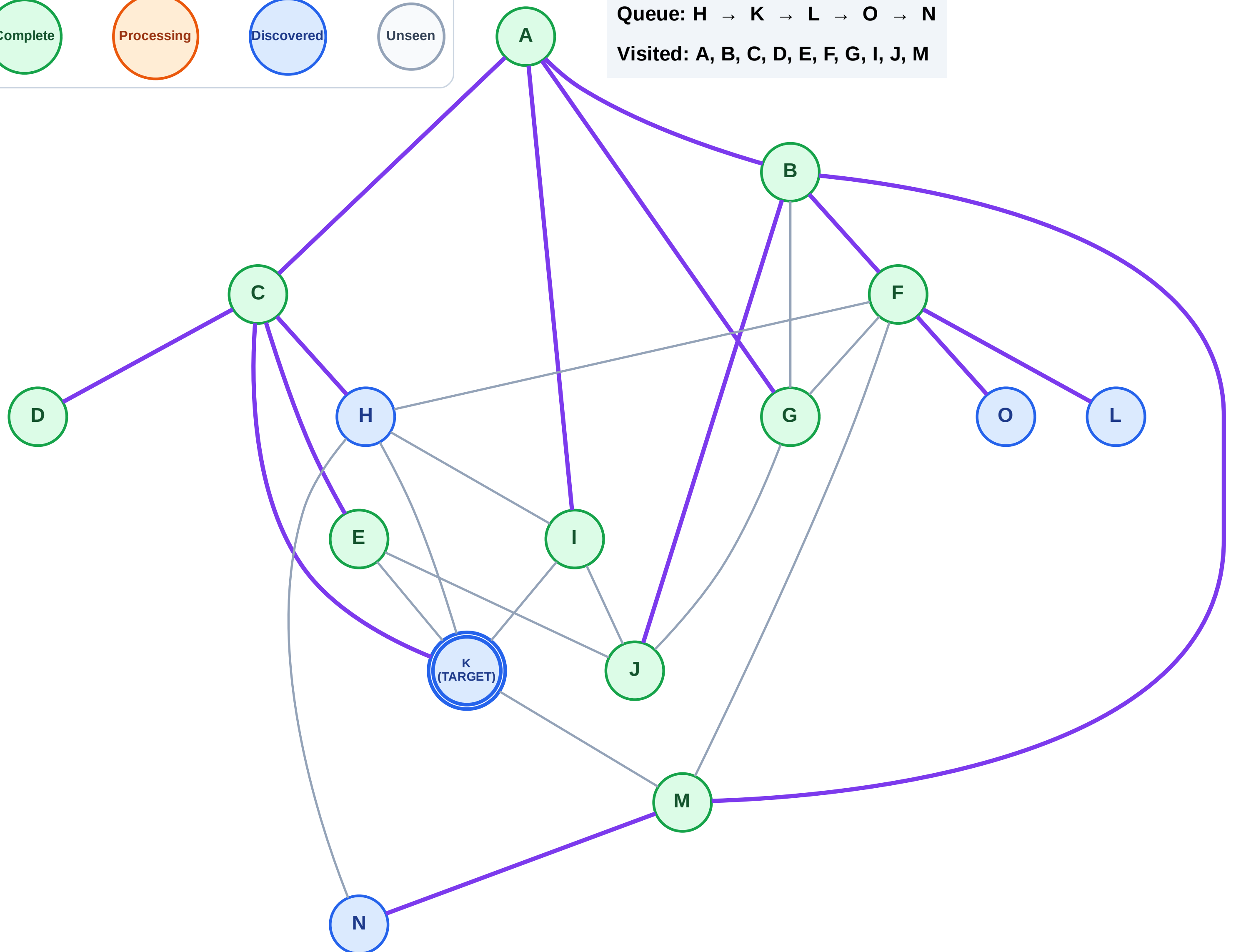
BFS Traversal

Step 21: No new neighbors from E



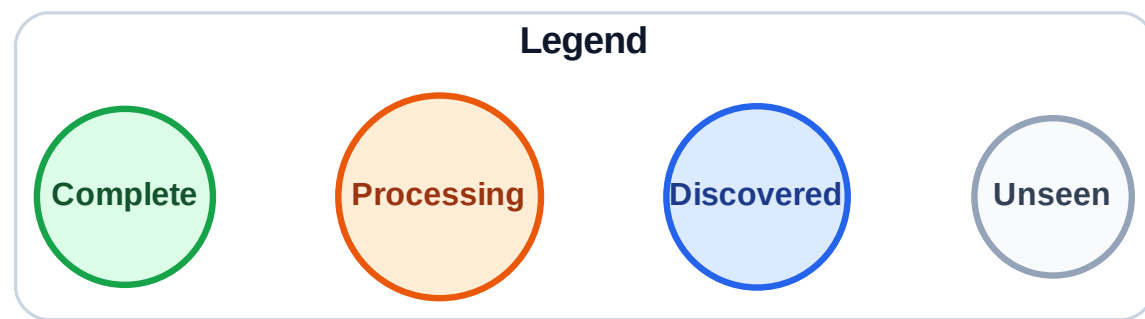
Queue: H → K → L → O → N

Visited: A, B, C, D, E, F, G, I, J, M



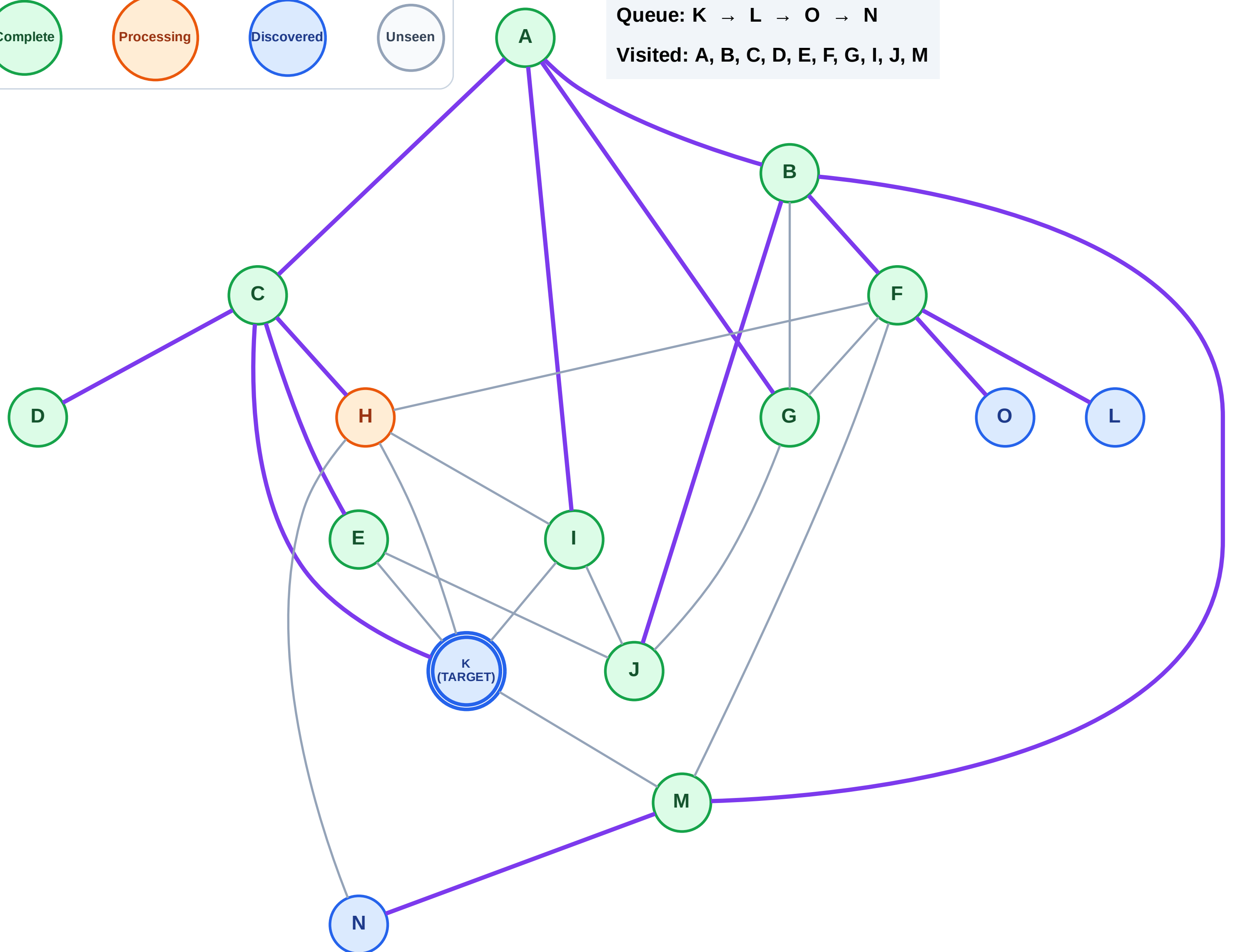
BFS Traversal

Step 22: Process node H



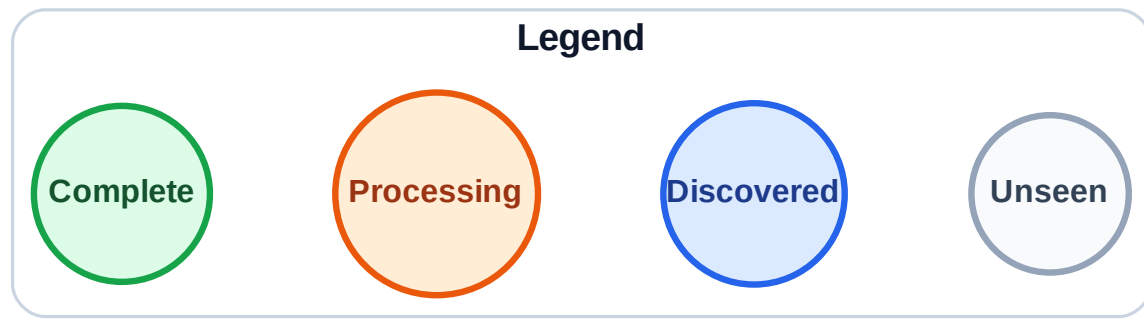
Queue: K → L → O → N

Visited: A, B, C, D, E, F, G, I, J, M



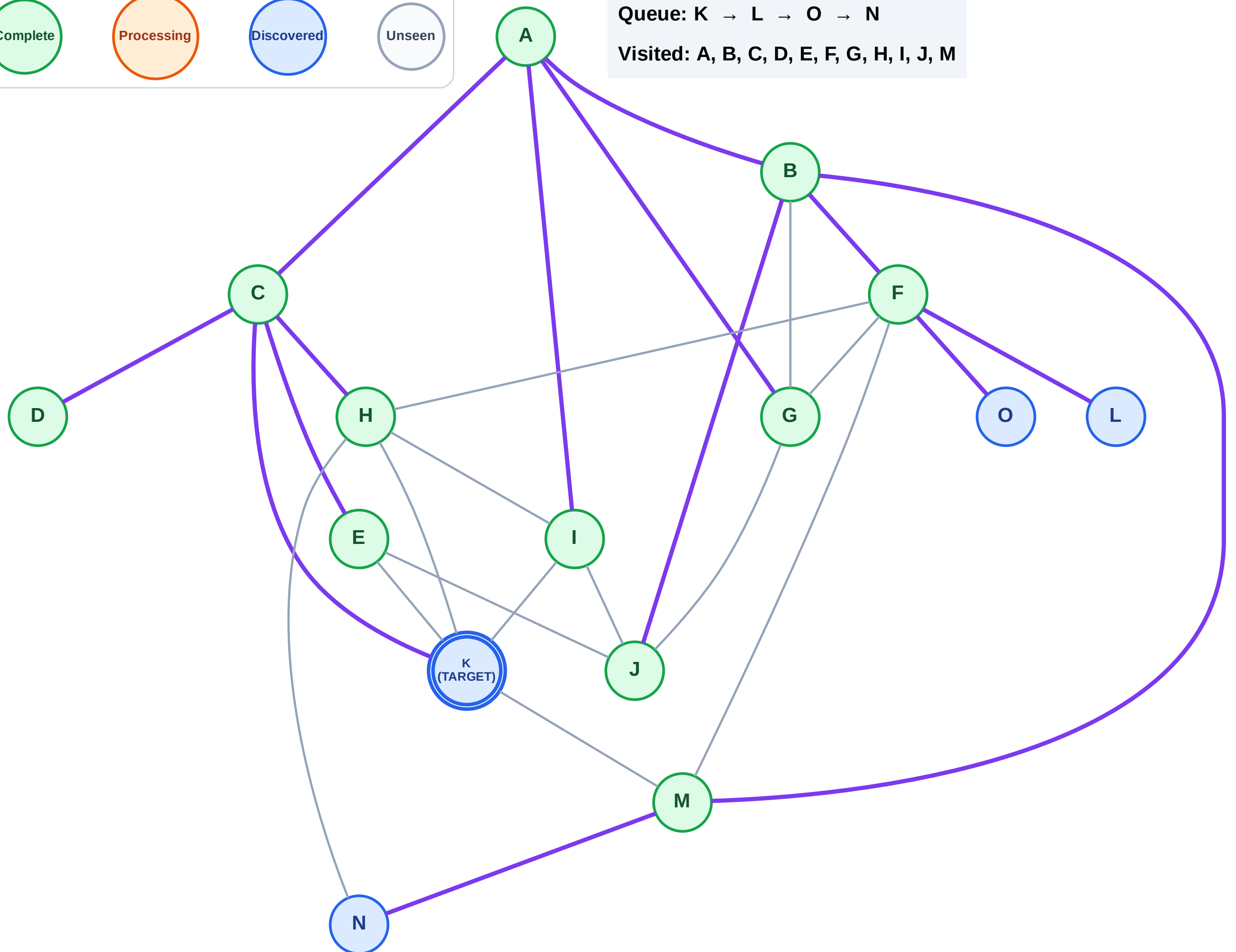
BFS Traversal

Step 23: No new neighbors from H



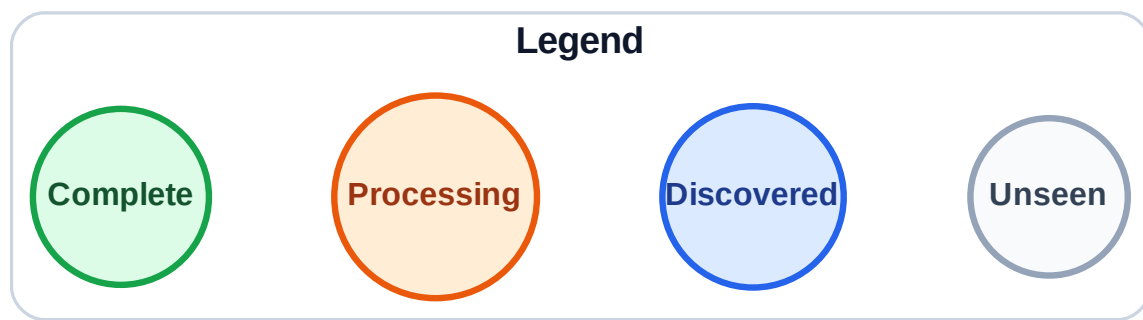
Queue: K → L → O → N

Visited: A, B, C, D, E, F, G, H, I, J, M

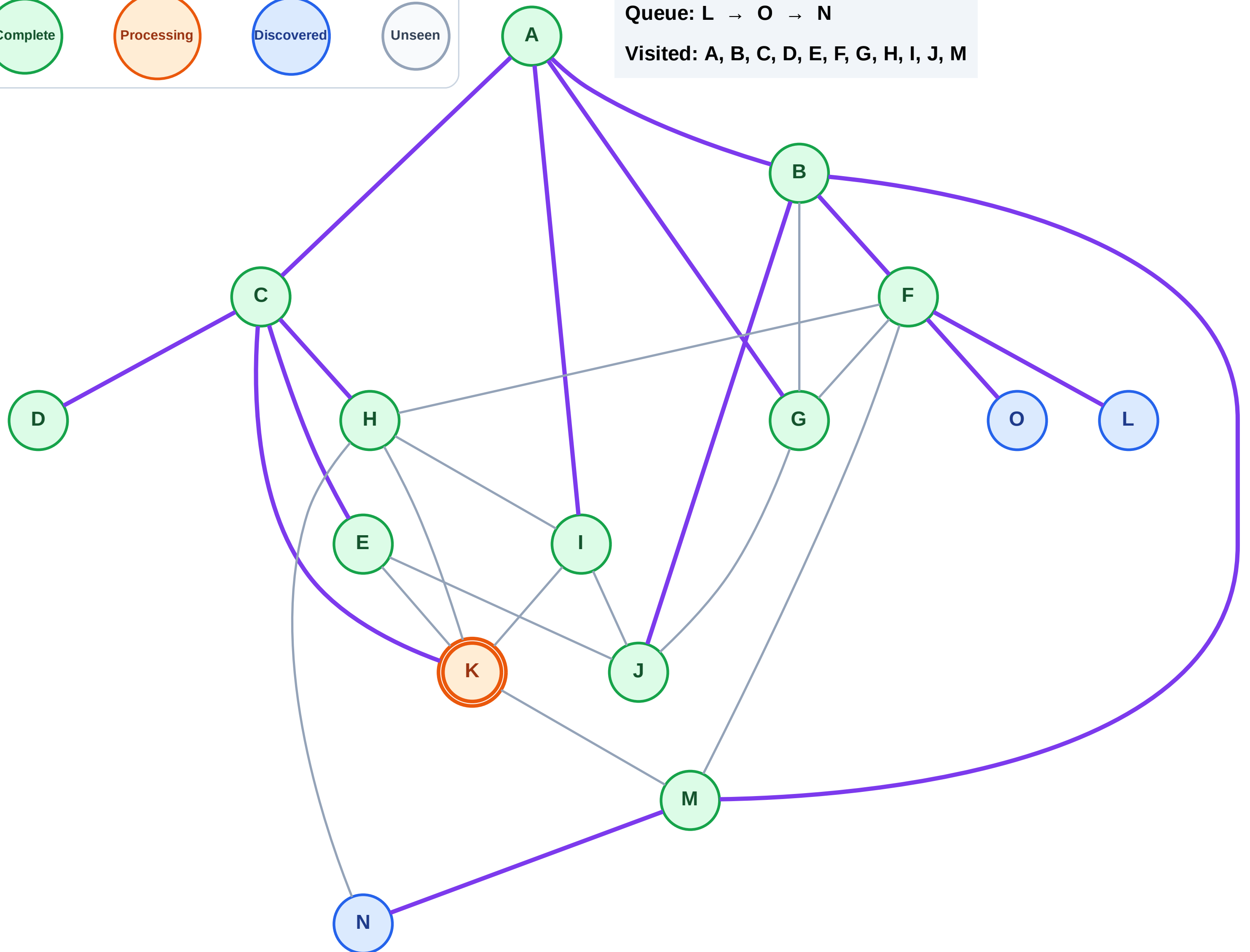


BFS Traversal

Step 24: Process node K



Queue: L → O → N
Visited: A, B, C, D, E, F, G, H, I, J, M



BFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend

Complete

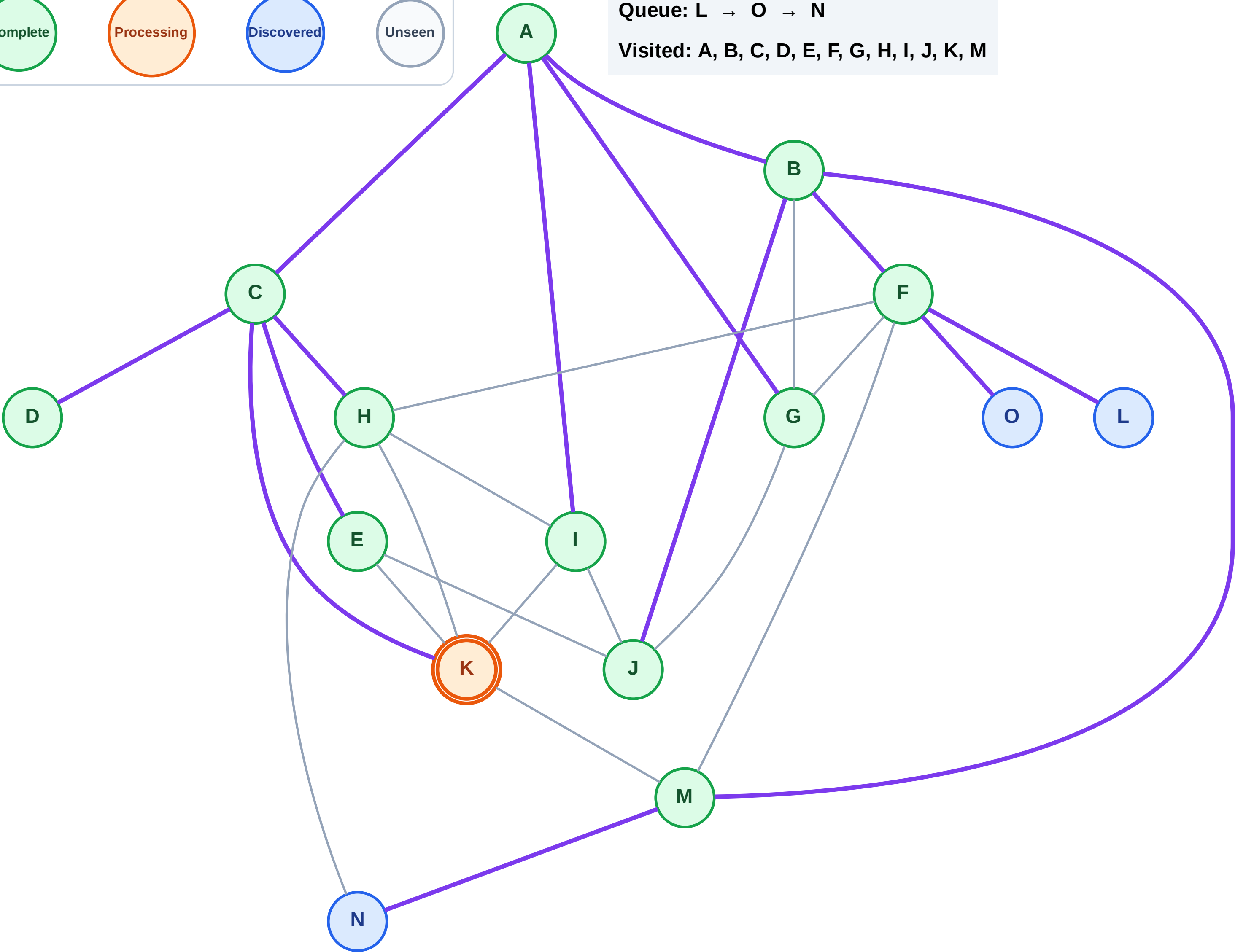
Processing

Discovered

Unseen

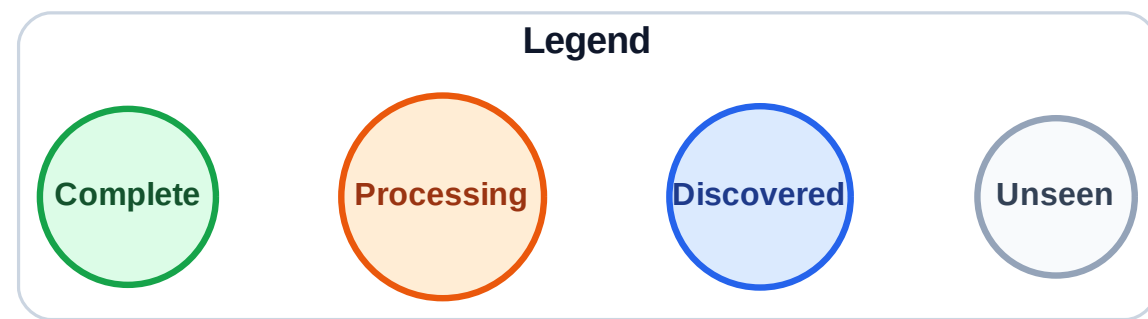
Queue: L → O → N

Visited: A, B, C, D, E, F, G, H, I, J, K, M



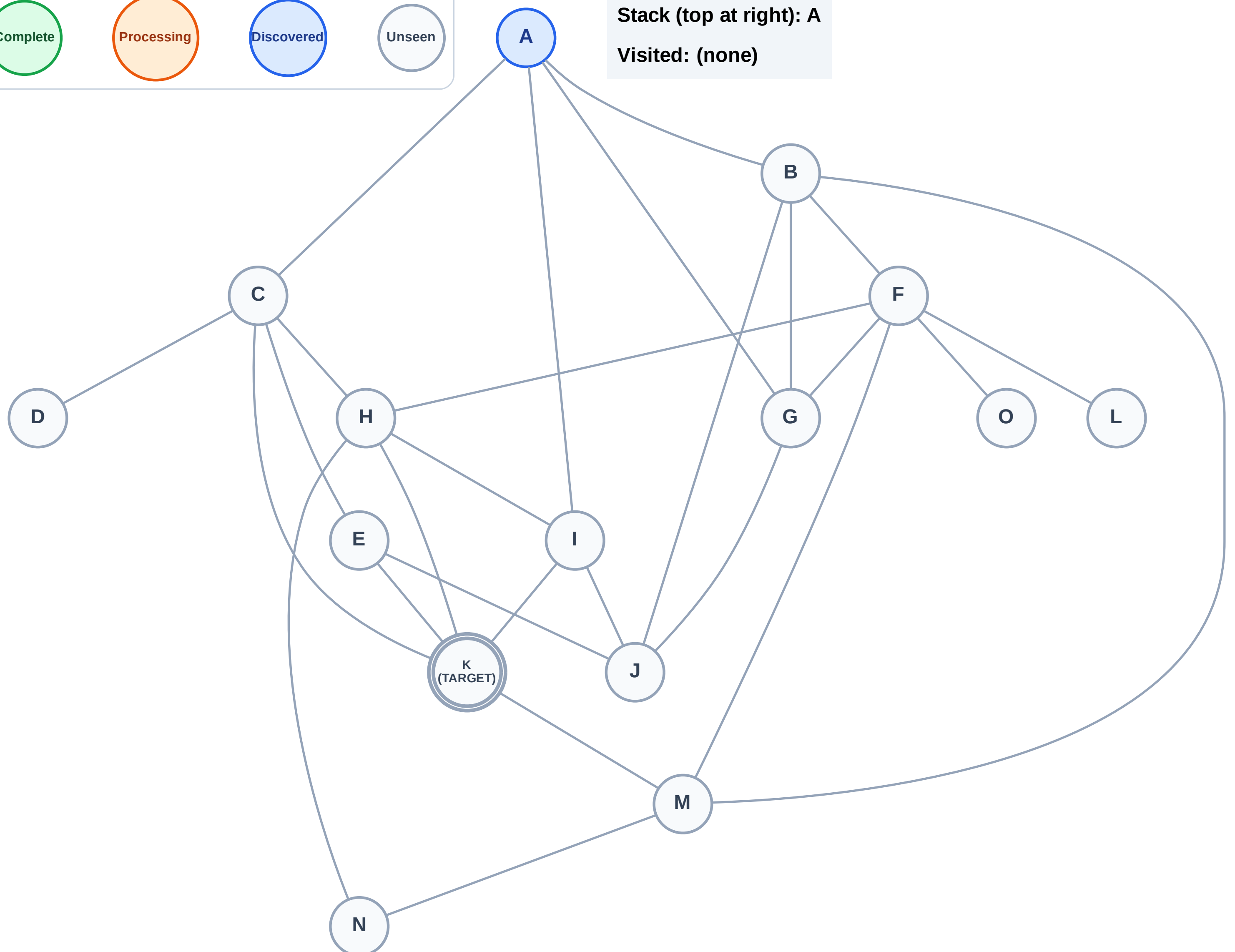
DFS Traversal

Step 1: Start at A (Seeking target node K)



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

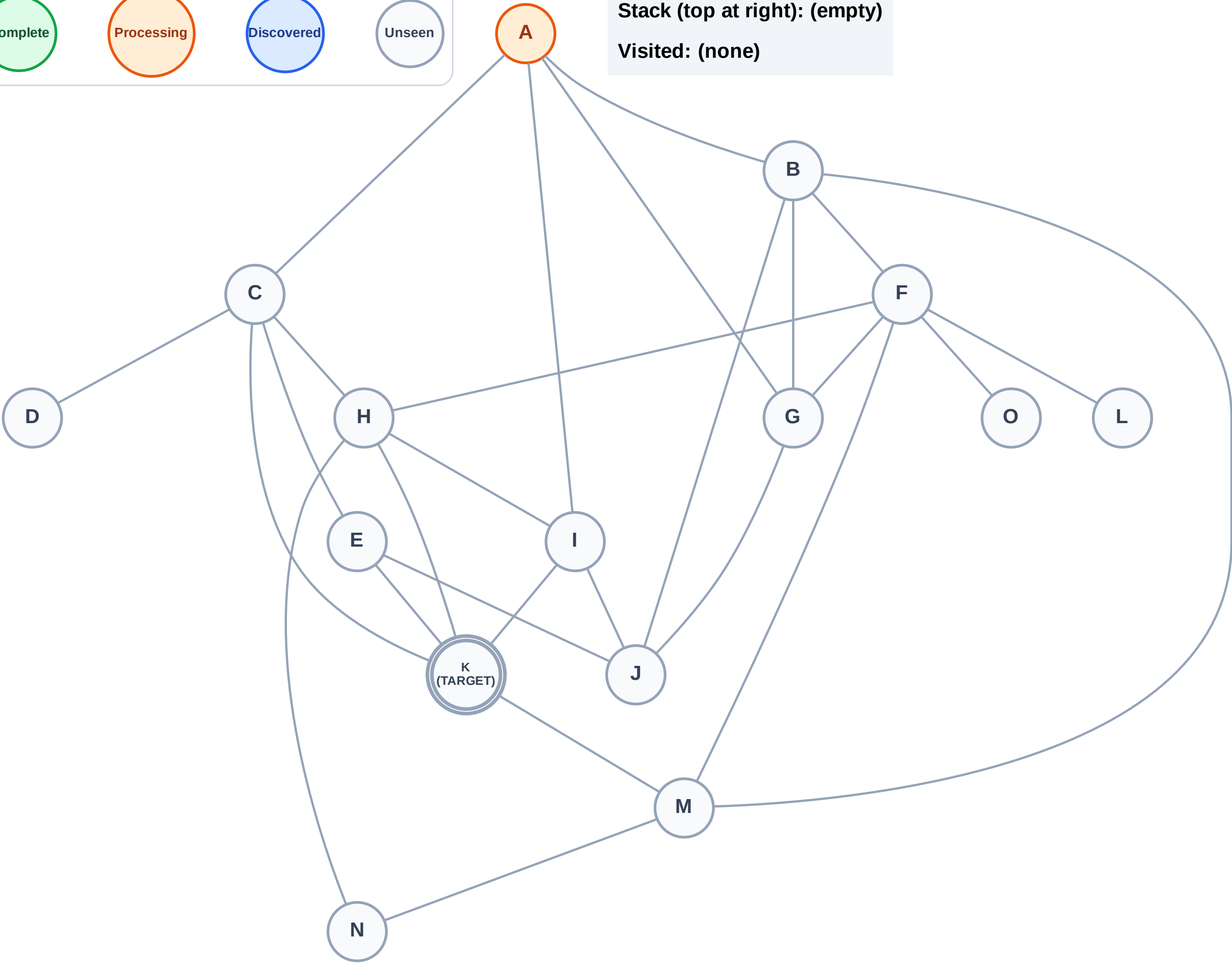
Complete

Processing

Discovered

Unseen

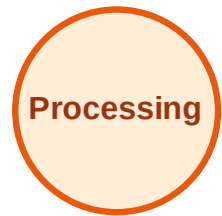
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

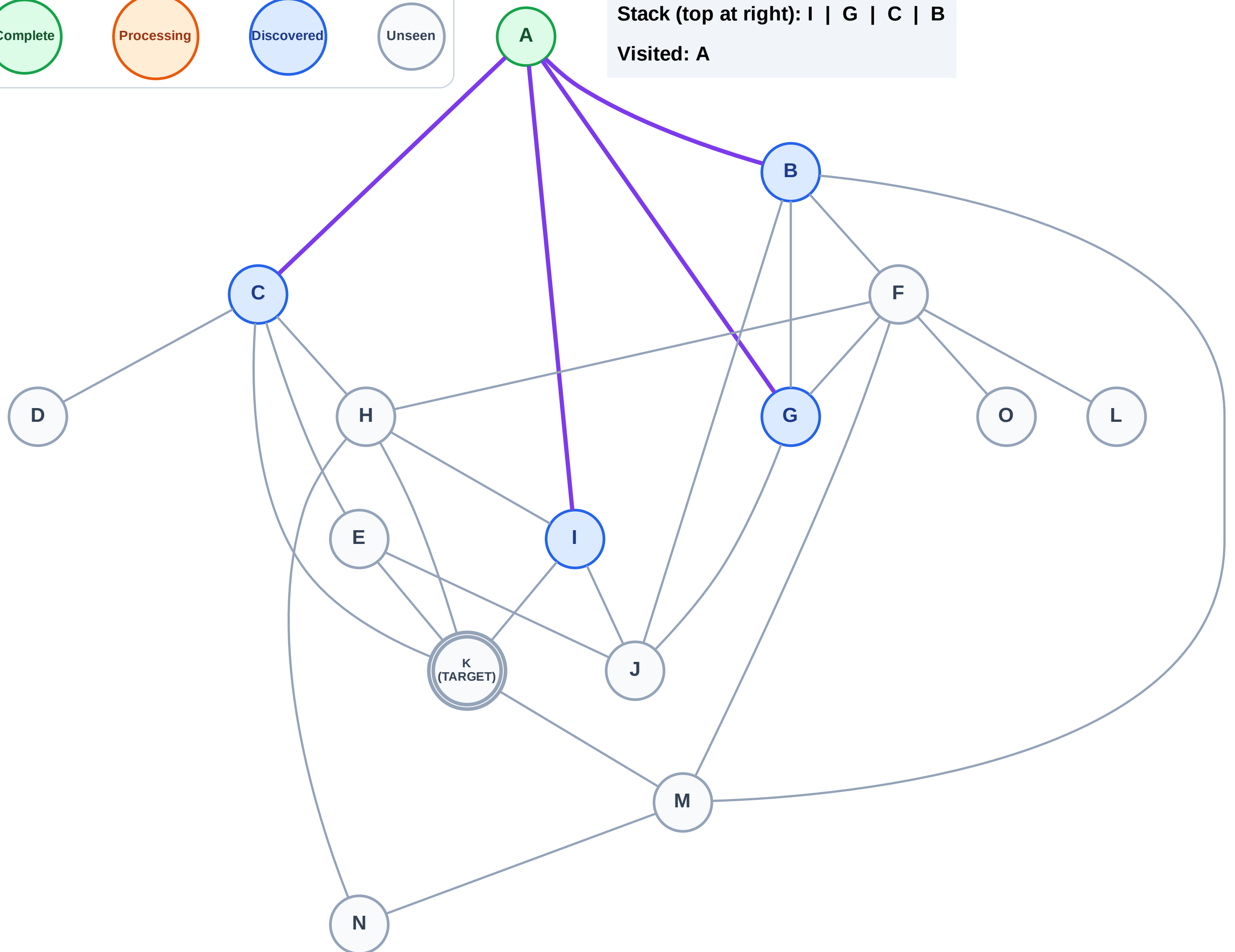
Step 3: Discover I, G, C, B from A

Legend



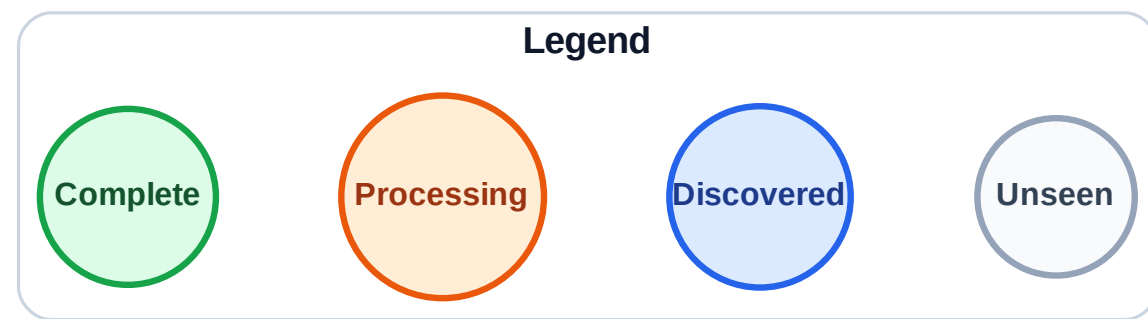
Stack (top at right): I | G | C | B

Visited: A



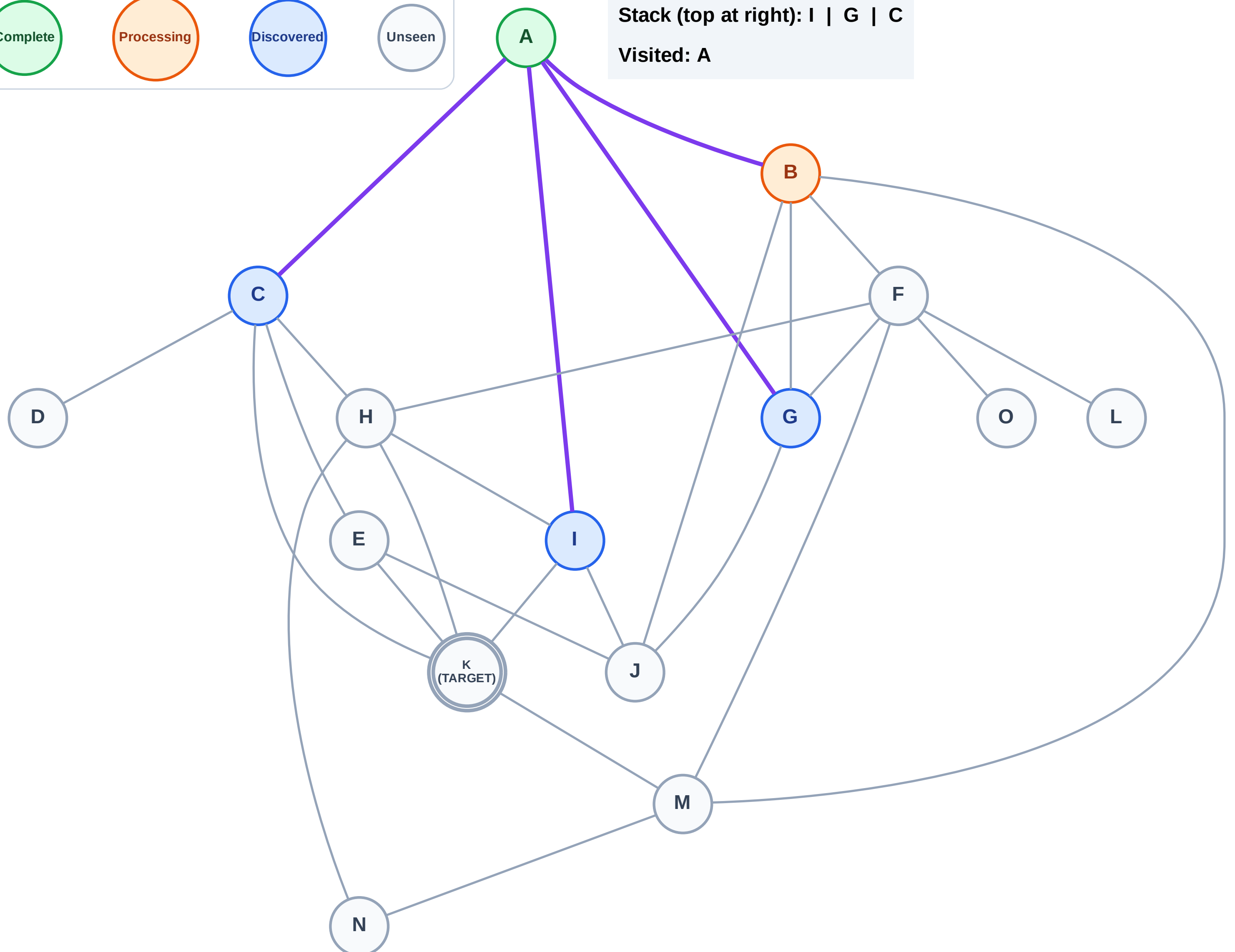
DFS Traversal

Step 4: Process node B



Stack (top at right): I | G | C

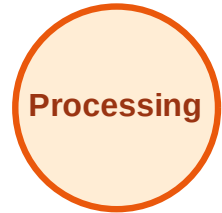
Visited: A



DFS Traversal

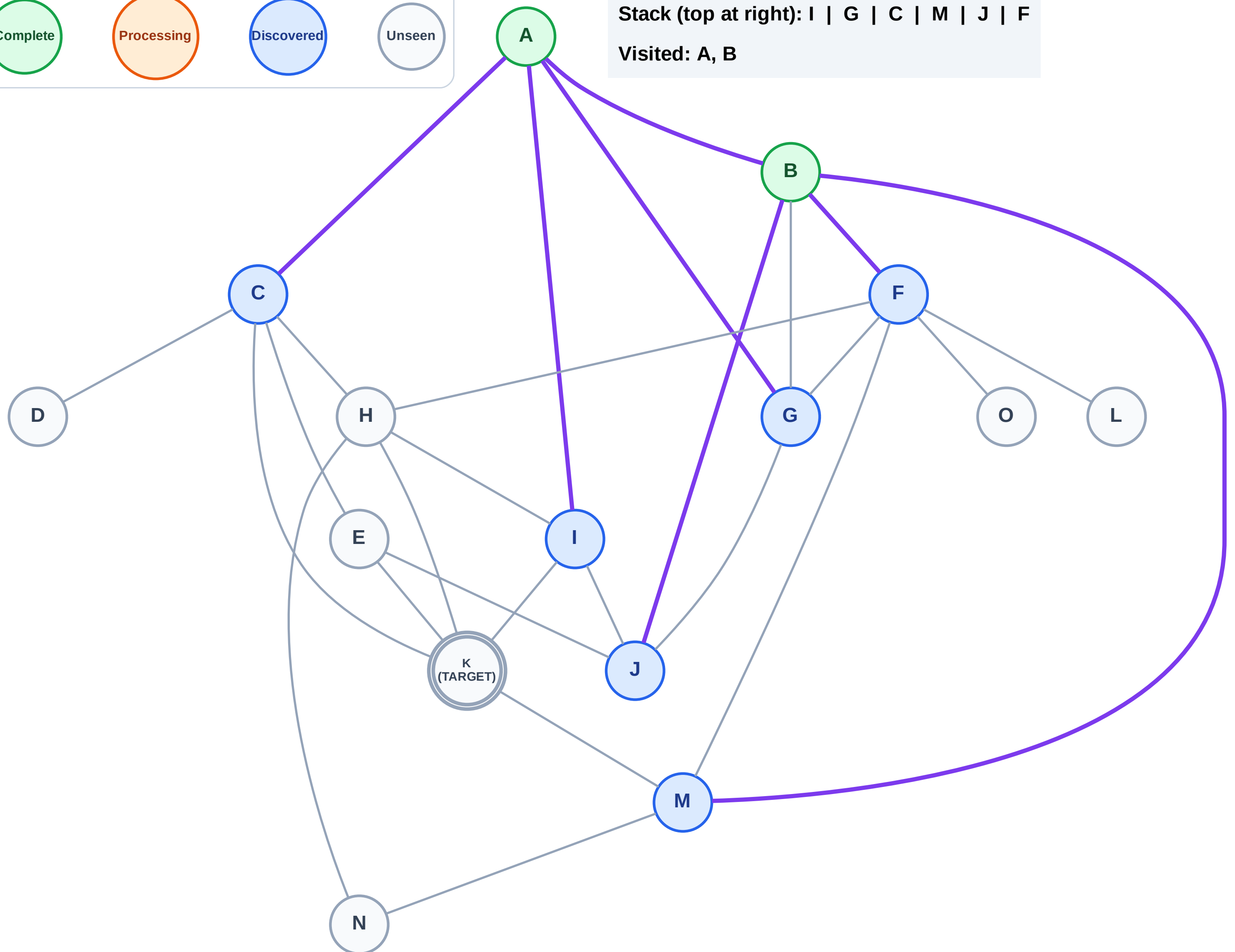
Step 5: Discover M, J, F from B

Legend



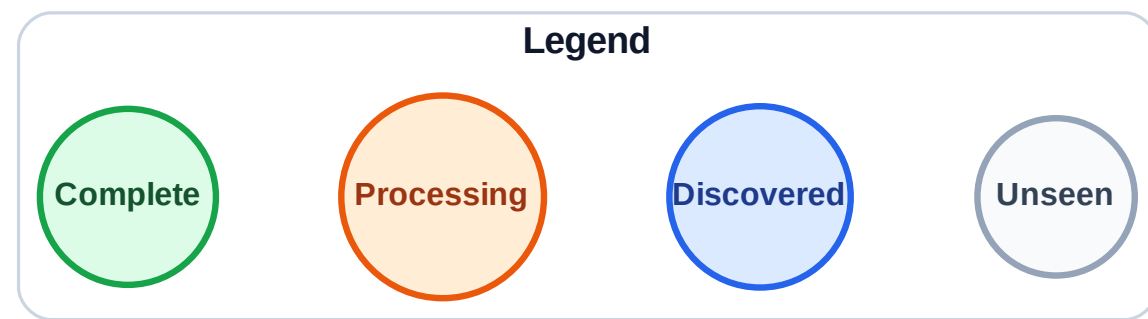
Stack (top at right): I | G | C | M | J | F

Visited: A, B



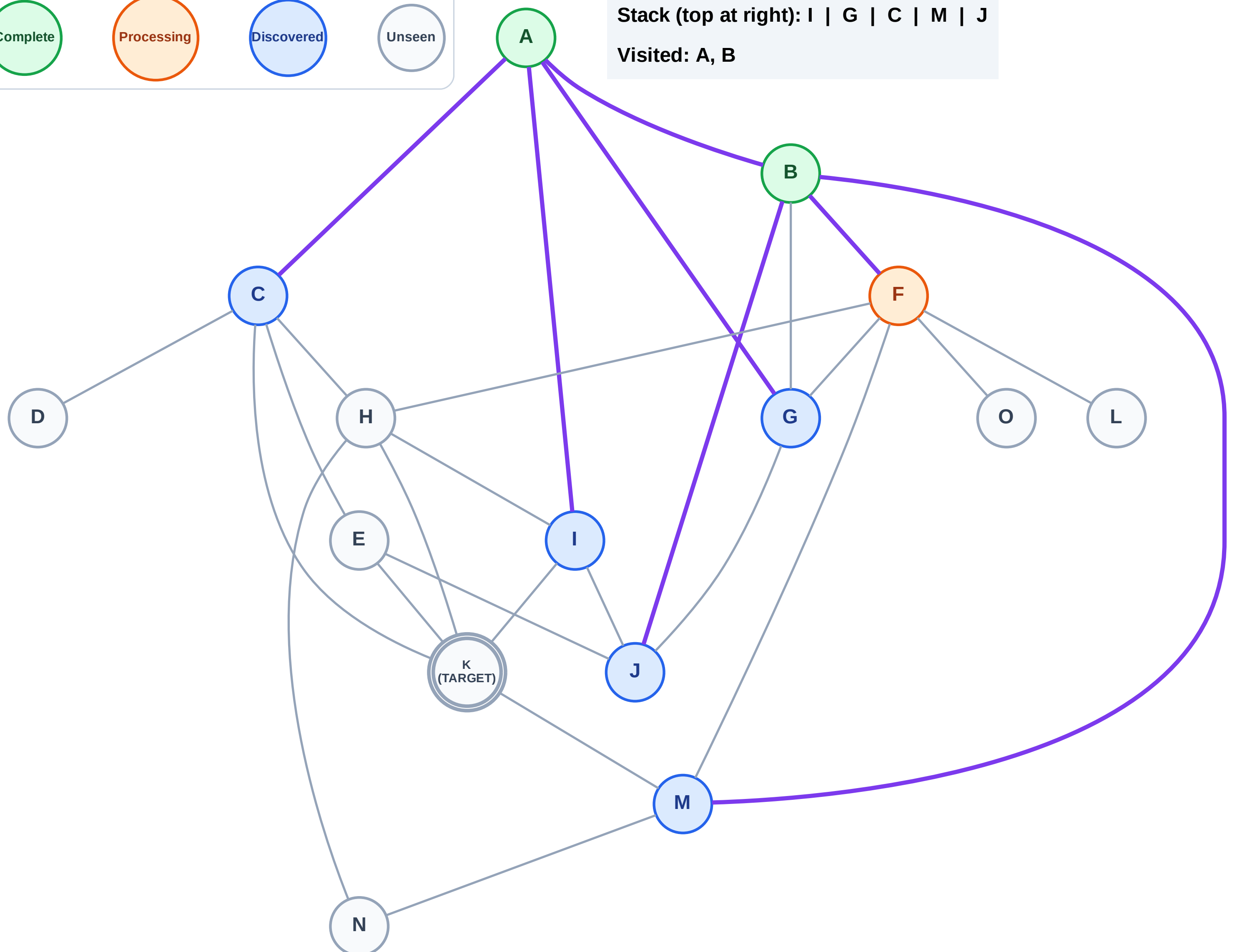
DFS Traversal

Step 6: Process node F



Stack (top at right): I | G | C | M | J

Visited: A, B

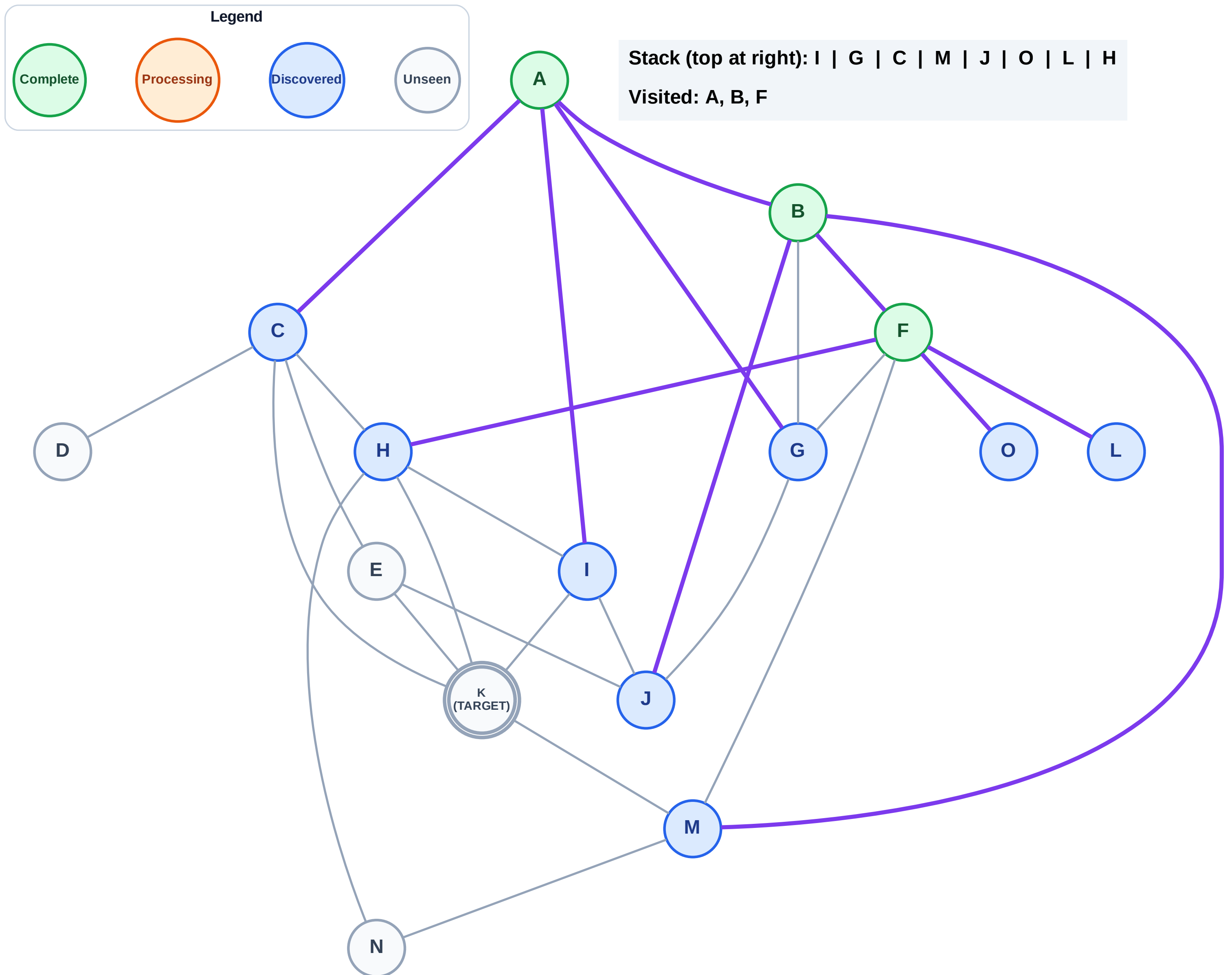


DFS Traversal

Step 7: Discover O, L, H from F

Stack (top at right): I | G | C | M | J | O | L | H

Visited: A, B, F



DFS Traversal

Step 8: Process node H

Legend

Complete

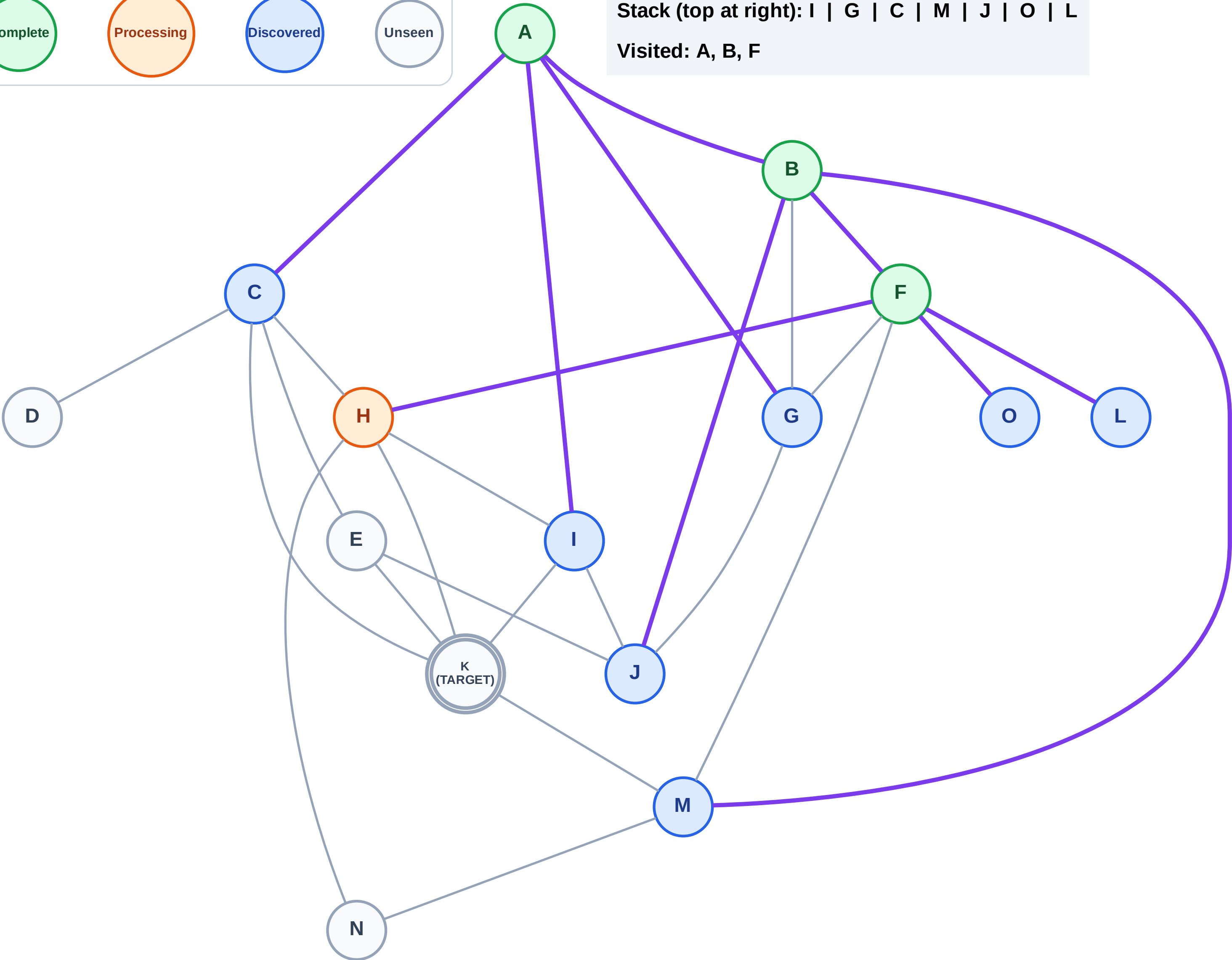
Processing

Discovered

Unseen

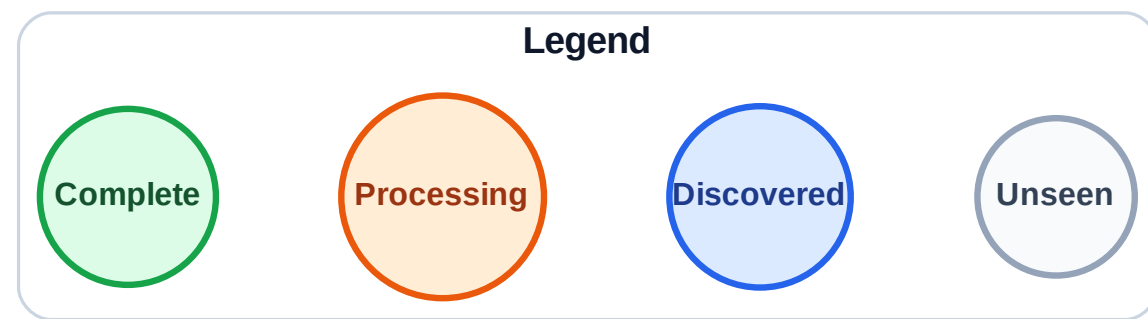
Stack (top at right): I | G | C | M | J | O | L

Visited: A, B, F



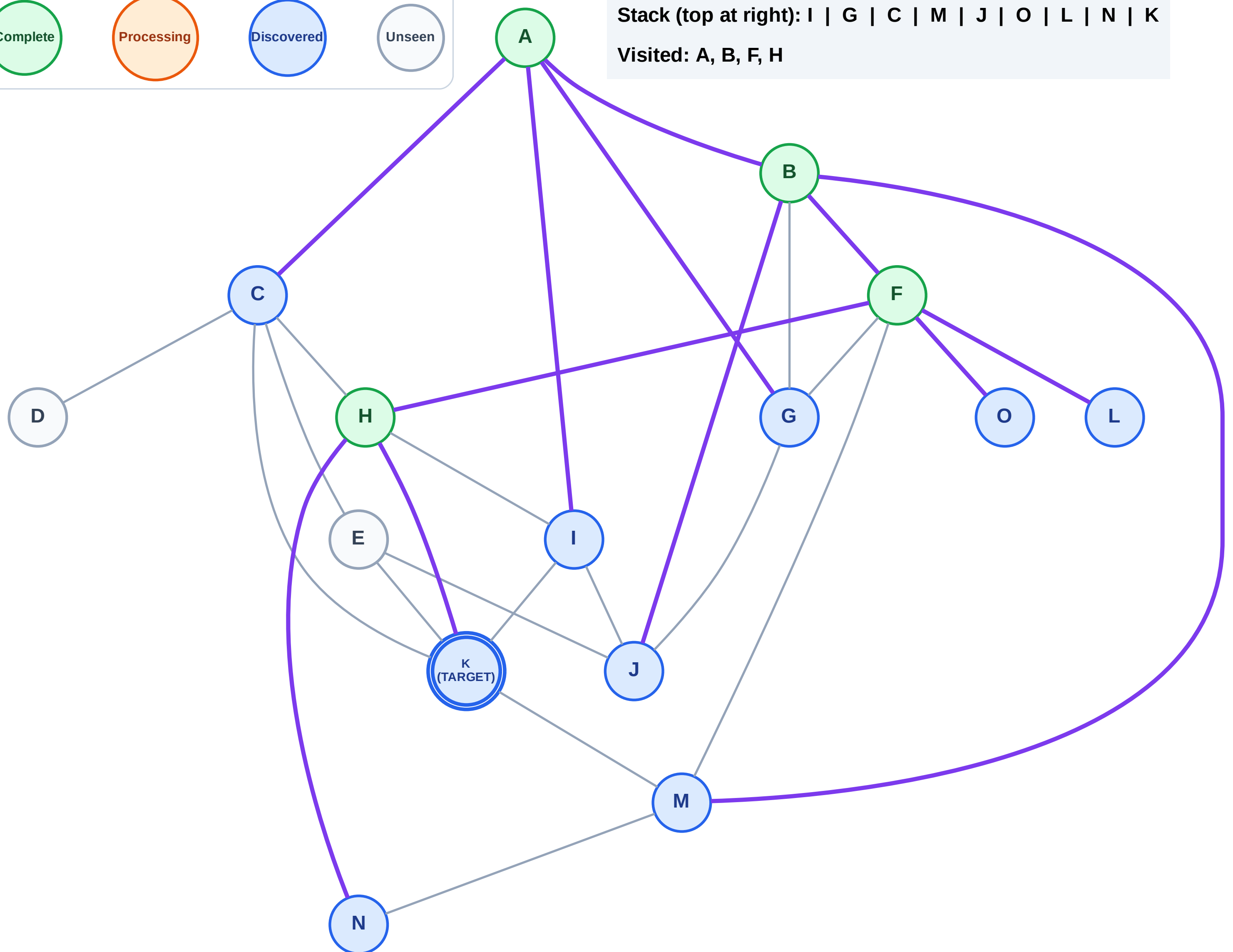
DFS Traversal

Step 9: Discover N, K from H



Stack (top at right): I | G | C | M | J | O | L | N | K

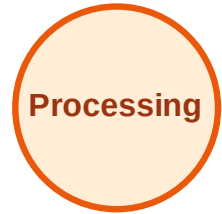
Visited: A, B, F, H



DFS Traversal

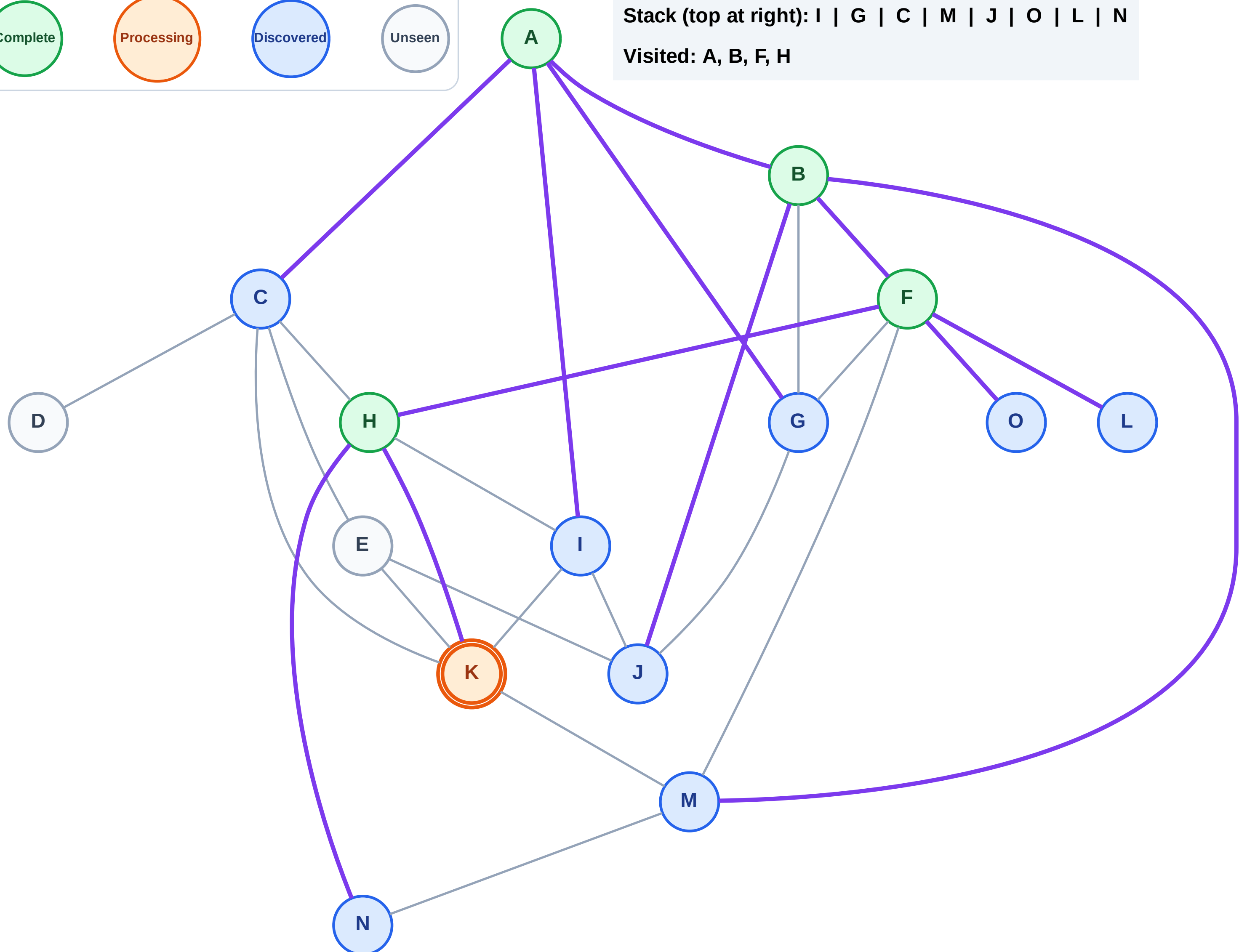
Step 10: Process node K

Legend



Stack (top at right): I | G | C | M | J | O | L | N

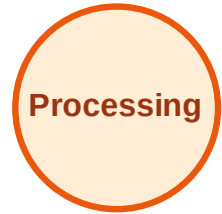
Visited: A, B, F, H



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (4 hops): A → B → F → H → K

Legend



Stack (top at right): I | G | C | M | J | O | L | N

Visited: A, B, F, H, K

